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Organizational Form and Trade Liberalization: Plant-Level Evidence

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Abstract. This paper studies how firms' internal organization shapes the impact of international trade. Using establishment-level data from the U.S. Census and a difference-in-difference specification, I find that, relative to standalone firms, conglomerates are more likely to restructure after trade liberalization episodes, focusing on their core competency and improving firm productivity and product market performance. Adjustments through the extensive margin account for the majority of the productivity growth differential between conglomerates and standalones experiencing trade shocks. Aggregate industry productivity remains relatively unchanged in industries dominated by conglomerates' core business but decreases significantly in others. My findings suggest that firms' internal organization has important consequences on the effects of trade policies.

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Keywords: firm organization • corporate restructuring • trade • aggregate productivity

1. Introduction

Globalization and trade liberalization have taken center stage in world economics over the last four decades. The United States alone saw the value of its imports increase more than 50 times, from \$40 billion dollars in 1970 to \$2,273 billion in 2015.¹ Although the existing literature has extensively studied how trade liberalization impacts aggregate industry productivity and economic growth,² limited evidence exists on the role of firms in international trade, particularly how "specialization and reallocation within the firm may turn out to play an important role in enhancing productivity and realizing welfare gains from trade," as noted by Bernard et al. (2007).

This paper directly contributes to this debate by focusing on how heterogeneity in firms' internal organization—operating in one (i.e., focused firms) versus more industries (i.e., conglomerates)—shapes the impact of trade liberalization. To capture major episodes of trade liberalization, I exploit a series of large declines in (U.S.) import tariff rates. By reducing the costs of entry of foreign firms into the U.S. market, the removal of trade barriers significantly increases the expected level of competition faced by import-competing firms. Importantly, these competitive threats cannot be perfectly anticipated and often are followed by sizable increases in ex post imports (Fresard 2010).

From a theoretical perspective, the interaction between firm organization and trade liberalization is

unclear. One issue is that resource allocation within a conglomerate based on the winner-picking hypothesis (e.g., Williamson 1975, Stein 1997) and the cross-subsidization view (e.g., Rajan et al. 2000) in their traditional form is only determined by an industry's short-term investment opportunity, which disregards the heterogeneity in segment importance or any strategic motivation to protect conglomerates' core competency. Therefore, I adopt a more nuanced approach and develop testable hypotheses that take into account the aggregate effects of firm actions.

Specifically, on one hand, I argue that the winner-picking hypothesis predicts that conglomerates can exploit their flexible organizational form to somehow mitigate any potential negative impact from the trade shocks. For instance, if conglomerates can do so either by improving the productivity of continuing plants or by closing (opening) less (more) efficient plants, one should expect a smaller productivity loss in the affected segments. On the other hand, the cross-subsidization view contends that, because conglomerate firms are more severely plagued by agency issues, such suboptimality might be exacerbated during a negative shock, leading to a greater productivity and performance loss for firms affected by the import tariff reductions.

Furthermore, to the extent that conglomerates' core businesses are often more efficient compared with their peripheral operations (Maksimovic and Phillips 2001) and are often associated with their corporate reputation (Atkins et al. 2006) or most popular products

(Fombrun and Gardberg 2000), it is reasonable to expect that, in the context of import tariff shocks, conglomerate firms might choose to refocus on their core business. The winner-picking hypothesis would also predict that conglomerates' actions together should improve firm efficiency and product market performance.

I test this series of hypotheses, which are fully developed in Section 2, by utilizing a difference-in-differences research design in which the treatment and control groups consist of firms that operate in multiple industries (i.e., conglomerates) and a single business segment (i.e., focused firms), respectively. Using a longitudinal panel of approximately 871,000 plant-year observations in the import-competing manufacturing sector between 1976 and 2004 constructed from micro-level data of the U.S. Census Bureau,³ I examine how conglomerates and focused firms differentially respond to trade shocks. The refined micro data allows me to not only track actions at plants that continue to exist but also restructuring actions through the extensive margin (e.g., plant opening, closure, divestiture, and acquisition).⁴

I begin my analysis with the intensive margin effects. I compare the changes in total factor productivity (TFP) in the three years after a trade shock for conglomerate plants and focused plants that are affected by these trade shocks. I find that treated conglomerate plants experience a significantly smaller (2.6 log points) drop in TFP compared with their focused counterparts, suggesting that continuing plants of conglomerate firms are better able to cope with the import competition. This result is not driven by conglomerate plants being, on average, older and larger than focused plants and is robust to controlling for time-invariant plant characteristics (i.e., plant fixed effects) as well as local economic shocks (i.e., state-year fixed effects).

I next turn to the extensive margin and examine how firms' propensity to restructure assets in industries affected by the trade shock changes and how such change varies across firms of different organizational forms. I find that conglomerates experiencing trade shocks are much more likely than their focused peers to open as well as close plants in the affected industries despite their lower probability of restructuring during other times (i.e., absent trade shocks). The economic magnitudes of the difference in restructuring likelihood between conglomerates and stand-alones are nontrivial. For instance, the likelihood for conglomerates to close down a plant affected by the shock increases by approximately 117% versus a 31% increase for focused firms.

To further shed light on the exact mechanism of restructuring, I investigate whether conglomerates' response depends on whether their core business segment is affected. The previous theoretical arguments suggest that conglomerate firms could choose

to refocus on their core competency to mitigate the negative impact of the import tariff shocks and to improve their long-term prospects. To this end, I reestimate the restructuring tests for the conglomerate subsample but partition plants into two groups: the core business and the peripheral operations. The results reveal both direct and indirect effects. In the direct effect, treated core plants experience slight change in the likelihood of closure, whereas their peripheral counterparts experience a significant increase. A strong and negative spillover effect exists for peripheral plants: they experience a significant increase in the probability of closure (8.1%) even when they themselves do not experience a shock but belong to a conglomerate firm that has some other plants affected by the import reduction. Given an average probability of plant closure of roughly 6.4%, this spillover effect represents a 127% increase in probability for peripheral operations. These results suggest that conglomerates' restructuring in the aftermath of the trade shock is indeed their attempt to focus on their core competency by shedding peripheral businesses and building more presence in the core industries.

Importantly, conglomerates' restructuring seems to reflect their winner-picking incentives as they are more likely to close treated plants in the lower part of the TFP distribution and are less likely to shut down high-TFP plants, making their overall restructuring productivity-enhancing. Moreover, conglomerates' refocusing on their core competency is also accompanied by consequences in the product market competition and aggregate industry productivity; in the three years after a trade shock, treated conglomerates gain market share and grow sales relative to their focused counterparts. In terms of aggregate industry productivity, those industries affected by trade shocks with a significant presence of conglomerates' core businesses experience a small productivity change, and other industries suffer from significant productivity losses.

To understand the relative economic magnitude of the extensive margin effects relative to the intensive margin effects,⁵ I compare productivity changes in the aftermath of trade shocks between a group of conglomerate plants and a matched sample consisting of focused plants within the same industry and similar in size at the onset of the shock. This quantification exercise, also employed by Davis et al. (2014) suggests that plant openings and closures account for approximately 78% of the firm-level differential in TFP change posttrade shocks, providing some of the first quantification on the relative importance of the intensive versus the extensive margin in firms' internal capital market.

Finally, I conduct additional tests to mitigate the possibility that the increased likelihood of corporate

restructuring documented so far is simply capturing a time trend in industry restructuring. That is, firms could engage in restructuring activities for reasons unrelated to trade shocks. In an attempt to rule out such a potential issue, I create placebo trade shocks by assigning fake treatments randomly to industries in years during which large tariff reductions actually occurred (in other industries). I find no significant relation between these placebo trade shocks and firms' ex post propensity to restructure. A general omitted time trend, therefore, appears unable to explain the observed results.

I also conduct several other checks to ensure that the main results are robust to alternative methods of estimating the total factor productivity, including the Olley and Pakes (1996) structural approach as well as using other performance measures (i.e., return on capital and operating margin) that are free of assumptions of the production function form. The results are unchanged by these alternative considerations.

This study contributes to two strands of literature. By studying the interaction between firms' internal organization and trade liberalization, it contributes to an extensive literature on competitive trade shocks. Although early literature concentrates on the impact of international trade on macroeconomic investment and employment,⁶ beginning with Bernard et al. (1995), recent studies provide evidence that firms play an important role in international trade (e.g., Tybout 2002, Bernard et al. 2007). This paper provides direct evidence on how heterogeneity in the scope of firms' operations can lead to differential impacts of trade liberalization on product market performance and aggregate industry productivity, highlighting the importance of considering firm boundaries when designing trade policy.

This paper also joins a large body of work on corporate diversification and internal capital markets. To date, the literature has found evidence on the impact of organizational form on a variety of corporate outcomes. For example, compared with focused firms, conglomerates tend to trade at a discount (Lang and Stulz 1994, Berger and Ofek 1995,⁷ Denis and Thothadri 1999), exhibit different investment behavior (Lamont 1997, Shin and Stulz 1998, Ahn et al. 2006, Seru 2014), and financial policies (Duchin 2010, Hann et al. 2013). A recent paper by Giroud and Mueller (2015) exploits plant-level data to examine the reallocation of resources, such as labor and capital, in multiestablishment firms when new airline routes are introduced between the headquarters and the plants and find that such reallocation exists only for financially constrained firms.⁸ This paper builds on this literature and offers consistent evidence on how firms' internal organization shapes the impact of trade liberalization and suggests

that the plausible mechanism is conglomerates' tendency to focus on and expand on their main competency.⁹

The remainder of this paper is organized as follows: Section 2 briefly reviews the theoretical background and develops the testable hypotheses. Section 3 discusses the data and the empirical methodology. Section 4 contains the empirical results. Section 5 discusses some caveats in interpreting the results and Section 6 concludes.

2. Hypothesis Development

2.1. Predictions Based on Traditional Theories

How would a firm's internal organizational form affect its ability to respond to a negative shock? Traditional theories offer opposite predictions on this issue. On one hand, the neoclassical view contends that conglomerate firms, because of their operations across multiple industries, can channel resources to segments with the best investment prospects (e.g., Williamson 1975, Stein 1997). Such an ability to pick the winners is particularly valuable when one or more segments of a conglomerate firm experience a negative shock, such as a sudden decrease in import tariffs. In particular, conglomerates might exploit their flexible structure to quickly draw physical capital away from the affected segments for which the expected investment opportunities worsen and redeploy them in other non-affected industries (Matsusaka and Nanda 2002). Recent research documents that human capital redeployment also constitutes an important channel through which firms' internal capital markets work (Tate and Yang 2015). Hence, compared with their focused counterparts, if conglomerate firms redeploy both the more productive physical capital and their human capital by moving them away from industries affected by the import shock, it would result in a more negative impact on overall productivity of the affected segments within a conglomerate firm. I summarize this in the following statement: conglomerates' segments affected by the import tariff shock experience a significantly greater productivity loss compared with their focused counterparts.

In direct contrast to the winner-picking view is the cross-subsidization hypothesis, which argues that conglomerate firms are plagued by agency problems and, thus, suboptimally invest in segments with unfavorable investment opportunities. Such distortions can arise when the divergence in resources and opportunities between different segments increases (Rajan et al. 2000) or when divisional managers engage in rent-seeking behavior (Scharfstein and Stein 2002). In the context of import tariff shocks, as the difference in investment opportunities between the affected segments and other segments increases, conglomerates might inefficiently invest more, instead of less, in the

affected segments. If reshuffling corporate resources entails reallocating more productive capital (both human and physical) toward the affected segments, it might result in an increase or, at the minimum, a smaller decrease in productivity among the affected segments,¹⁰ which leads to the following prediction: conglomerates' segments affected by the import tariff shock experience a smaller productivity loss compared with their focused counterparts.

2.2. Hypotheses with Heterogeneity in Segment Importance

2.2.1. Effect on Total Factor Productivity. Although the discussion in Section 2.1 provides a useful theoretical starting point for thinking about the implications of organizational form on firms' response, these traditional theories on corporate diversification treat all segments within a conglomerate firm homogeneously. In the context of import tariff shocks, this suggests that a conglomerate's decision to allocate (withdraw) resources to (from) a segment depends on only its investment opportunity irrespective of its importance for the firm overall.

In reality, although conglomerates operate in multiple lines of business, they are often organized around a core competency. In particular, conglomerates' core segments are often larger in size and significantly more efficient than their smaller noncore counterparts in peripheral industries (Maksimovic and Phillips 2001).¹¹ Matsusaka (2001) offers a theoretical rationale for why conglomerates' core segments might be more efficient. In his model, conglomerates have a firm-specific organizational ability that has different values in different industries, and firms experiment to find out whether their organizational ability is a good match to a specific industry. This experimental and evolutionary view leads to the conclusion that the match quality between firms' organizational/managerial capital and the underlying assets is the highest for the core segment of a conglomerate firm.

Furthermore, conglomerates' core business segment is of significant and often long-term strategic importance for these firms. For example, previous studies find that conglomerates' core business is often closely tied to their corporate reputation (Atkins et al. 2006) or most popular products (Fombrun and Gardberg 2000). It is, therefore, important to consider segment importance in formulating hypotheses on conglomerates' response to import tariff reductions.

As a result of these considerations, the optimality of conglomerates' response to trade shocks cannot be evaluated based solely on whether they channel *more* or *fewer* resources to an industry affected by the import tariff shock, which admittedly experiences a deterioration in its short-term investment opportunity. Rather, a more nuanced view that takes into

account the aggregate effects of firm actions is called for. Specifically, if conglomerates can take advantage of their more flexible (preexisting) organizational form to cope with a tariff shock, either by improving the productivity of continuing plants or by closing (opening) less (more) efficient plants, one should expect a smaller productivity loss in the affected segments.¹² I summarize this argument in the following hypothesis:

Hypothesis 1 (Winner-Picking Hypothesis). *Conglomerates' segments affected by an import tariff shock experience a smaller productivity loss compared with their focused counterparts.*

On the other hand, under the cross-subsidization view, conglomerates are plagued by more severe agency problems. In the context of tariff shocks, conglomerates might be unable to effectively and systematically reevaluate each segment's long-term investment potential, resulting in suboptimal resource (re)allocation, which manifests itself in, for instance, failing to close plants that are inefficient to operate, (inefficiently) terminating the more productive operations, or even opening plants that are inefficient. These inefficiencies together would imply that conglomerates cannot effectively respond to the intensified product market competition, and experience larger productivity losses:

Hypothesis 1a (Cross-Subsidization Hypothesis). *Conglomerates' segments affected by an import tariff shock experience a larger productivity loss compared with their focused counterparts.*

2.2.2. Importance of the Segments: Core vs. Noncore. As discussed in Section 2.2.1, traditional winner-picking and cross-subsidization hypotheses on corporate diversification disregard the heterogeneity in segment importance. Conglomerates' core segments have long-term strategic importance to the firms' product market success, corporate identity, survival, and prosperity. Therefore, to effectively deter entry or to respond more aggressively to potential competitors, conglomerates might build even more capacity, not less, in their core competency despite the short-term worsening investment prospects. Conglomerates' overinvestment in their core competency is similar to the intuition in a series of influential papers on strategic competition and entry deterrence (e.g., Spence 1977, Dixit 1979, Fudenberg and Tirole 1984).

These theories remain silent on the exact mechanism through which firms invest. Firms can alter their capital investment and employment decisions at existing plants (i.e., the intensive margin) and can also adjust through corporate restructuring by engaging in plant opening, closure, acquisition, and divestiture (i.e., the extensive margin). Because existing theories do not

address the exact form of adjustment, this is ultimately an empirical question. I state the hypotheses related to segment importance first in its null form, followed by the alternative hypothesis:

Hypothesis 2. *The importance of a segment within a conglomerate firm does not affect how the segment responds to the negative shock brought about by the import tariff reduction.*

Hypothesis 2a. *Conglomerate firms' response to the import tariff shock depends on the segments' importance. In particular, conglomerate firms might allocate more (fewer) resources (through intensive and/or extensive margin adjustments) to a core (peripheral) segment affected by the shock.*

2.2.3. Conglomerates' Response: Efficiency vs. Agency.

Despite a possible strategic reason to allocate more resources toward their core segments, conglomerates' response to the trade shocks can be efficiency-improving with a focus on long-term firm and industry prospects (i.e., winner-picking) or a further manifestation of their more severe agency problems. To clarify, I investigate whether conglomerates' resource allocation decisions are correlated with the segment's own productivity, the industry's forward-looking investment opportunities, and future product market performance and firm value, summarized as follows:

Hypothesis 3 (Winner-Picking Hypothesis). *Conglomerates shift resources into more productive plants and plants with better industry prospect, and, in the aggregate, improve firm value and product market performance.*

The cross-subsidization hypothesis would predict the opposite:

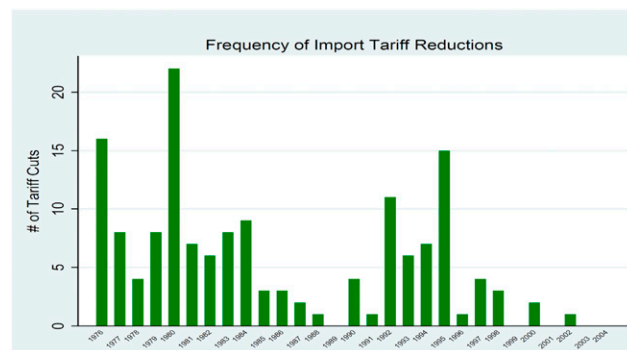
Hypothesis 3a (Cross-Subsidization Hypothesis). *Conglomerates shift resources into less productive plants and plants with worse industry prospects and, in the aggregate, reduce firm value and product market performance.*

3. Empirical Setting, Data, and Sample

3.1. The Empirical Setting

The main empirical setting in this paper centers around trade liberalization episodes characterized by large import tariff reductions depicted in Figure 1. The key identifying assumption is that these import tariff changes are reasonably exogenous to firms' organizational form. However, two potential issues could pose a challenge to the interpretation of my main results. First, import tariffs may be associated with past industry performance and conditions. For instance, import tariffs could be increased to protect infant industries (e.g., Hamilton 1791) and lowered when these industries become mature and competitive. Second, the change in import tariffs could be a

Figure 1. (Color online) Distribution of Trade Shocks over Time



consequence of firms' active lobbying (e.g., Lenway et al. 1996). If conglomerate firms, because of their larger size relative to focused firms, are more powerful lobbyists, they can better anticipate and prepare for reductions in import tariffs.¹³ I address these concerns next.

I examine whether past industry conditions and trends affect the likelihood of a trade shock. Following Fresard and Valta (2016), a four-digit Standard Industrial Classification (SIC) industry is defined as experiencing a trade shock when the following conditions are met: First, the annual negative change in import tariff rates in a given four-digit SIC industry has to exceed three times the median industry change during the sample period.¹⁴ Second, to avoid capturing episodes of transitory tariff reduction, I exclude cases in which the tariff reduction is followed by equivalently large increases in import tariff rates over the next three years (i.e., $t + 1$, $t + 2$, and $t + 3$). Finally, the absolute annual change in tariff has to be larger than 1%. At the four-digit SIC industry level, using various specifications (conditional logistic, probit, and linear probability model), I estimate whether lagged industry characteristics can predict the occurrence of a major trade shock. I include as covariates lagged values of the logarithm of the number of conglomerate firms in the industry, the logarithm of the total number of firms in the industry, industry output, industry Herfindahl index, change in industry output, and the percentage of industry output produced by conglomerate firms. Table 1 contains the results. If poor past industry performance is a stronger predictor for the likelihood of a trade shock, one would expect the coefficient on *Change in Industry Output* to be significantly negative across different specifications. If conglomerates lobby to influence the import tariffs, the coefficient on *Log Number of Conglomerate Firms* should be significant. However, as shown in Table 1, with the exception of the *Log Number of Firms*, none of the other independent variables is statistically significant. Admittedly, it is still possible to have

Table 1. Predictability of Trade Shocks

	Logit (1)	Probit (2)	LPM (3)
<i>Log No. of Conglomerate Firms</i> _{<i>t</i>−1}	0.159 (0.107)	0.085 (0.061)	0.009 (0.007)
<i>Log No. of Firms</i> _{<i>t</i>−1}	−0.088* (0.048)	−0.050* (0.027)	−0.006 (0.004)
<i>Industry Output</i> _{<i>t</i>−1}	−0.039 (0.069)	−0.017 (0.039)	−0.002 (0.004)
<i>Industry Herfindahl</i> _{<i>t</i>−1}	−1.473 (1.122)	−0.841 (0.620)	−0.060 (0.052)
<i>Change in Industry Output</i> _{<i>t</i>−1}	−0.012 (0.014)	−0.007 (0.009)	−0.001 (0.001)
% of Ind. Output by Conglomerates _{<i>t</i>−1}	−0.002 (0.001)	−0.001 (0.001)	0.0201 (0.019)
Year fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Observations	8,000	8,000	10,000
(Pseudo) <i>R</i> ²	0.16	0.15	0.07

Notes. This table regresses the tariff cut dummy on lagged industry-level variables. The tariff cut dummy takes a value of one if an industry experiences a tariff reduction that is three times larger than the mean change during the sample period. *Log No. of Conglomerate Firms* is the logarithm of the number of conglomerate firms in a four-digit SIC industry in a year. *Log No. of Firms* is logarithm of the number of firms in a four-digit SIC industry in a year. *Industry Output* is the logarithm of the total value of shipments in a four-digit SIC industry. *Industry Herfindahl* is the sum of squared percentage of sales in the four-digit SIC industry. *Change in Industry Output* is the annual percentage change in industry output, that is, $(\text{Industry Output}_t - \text{Industry Output}_{t-1}) / \text{Industry Output}_{t-1}$. % of Ind. Output by Conglomerates is the percentage of total value of shipments in a four-digit SIC industry that are produced by a conglomerate. All other variables are defined in the appendix. Heteroscedasticity robust standard errors are in parentheses and are corrected for clustering at the industry level.

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

reverse possibility: industries with trade shocks go through mergers, resulting in more conglomerates. Although lagging the independent variables can partially mitigate this issue, it cannot rule it out completely.

To further alleviate the concern of potential anticipation and to better understand the economic impact associated with these import tariff reductions, for each affected four-digit SIC industry, I plot the average number of closed and newly opened plants as well as those acquired and divested during the five-year window surrounding these tariff cuts. It is evident in Figure 2 that the import tariff shocks are followed by intense restructuring activities in the affected industries. For example, the average number of closed plants, which is approximately 43 each year prior to the shock, surges to almost 50 in the aftermath of a major tariff cut: a 16% increase. Meanwhile, the number of entrants experiences a sharp decline in the year of the shock and restabilizes at a lower equilibrium

level relative to the preshock era. Similarly, acquisition and divestiture also exhibit significant changes at the onset of the trade shock. Taken together, these patterns suggest it is unlikely that firms can perfectly anticipate the tariff cuts; otherwise, one would expect to see significant changes even before the shocks occur.

In Figure 3, I plot for each industry the average growth in sales, employment, capital investment, and productivity, and the results again demonstrate the dramatic changes taking place at the time of and immediately following the shock but with no discernible patterns prior to the treatment.

Although the figures in this section provide a general sense of the overall economic events around the import tariff shocks, these are nonparametric in nature and do not provide a rigorous analysis in a multivariate panel regression framework. I defer these tests to Section 4.

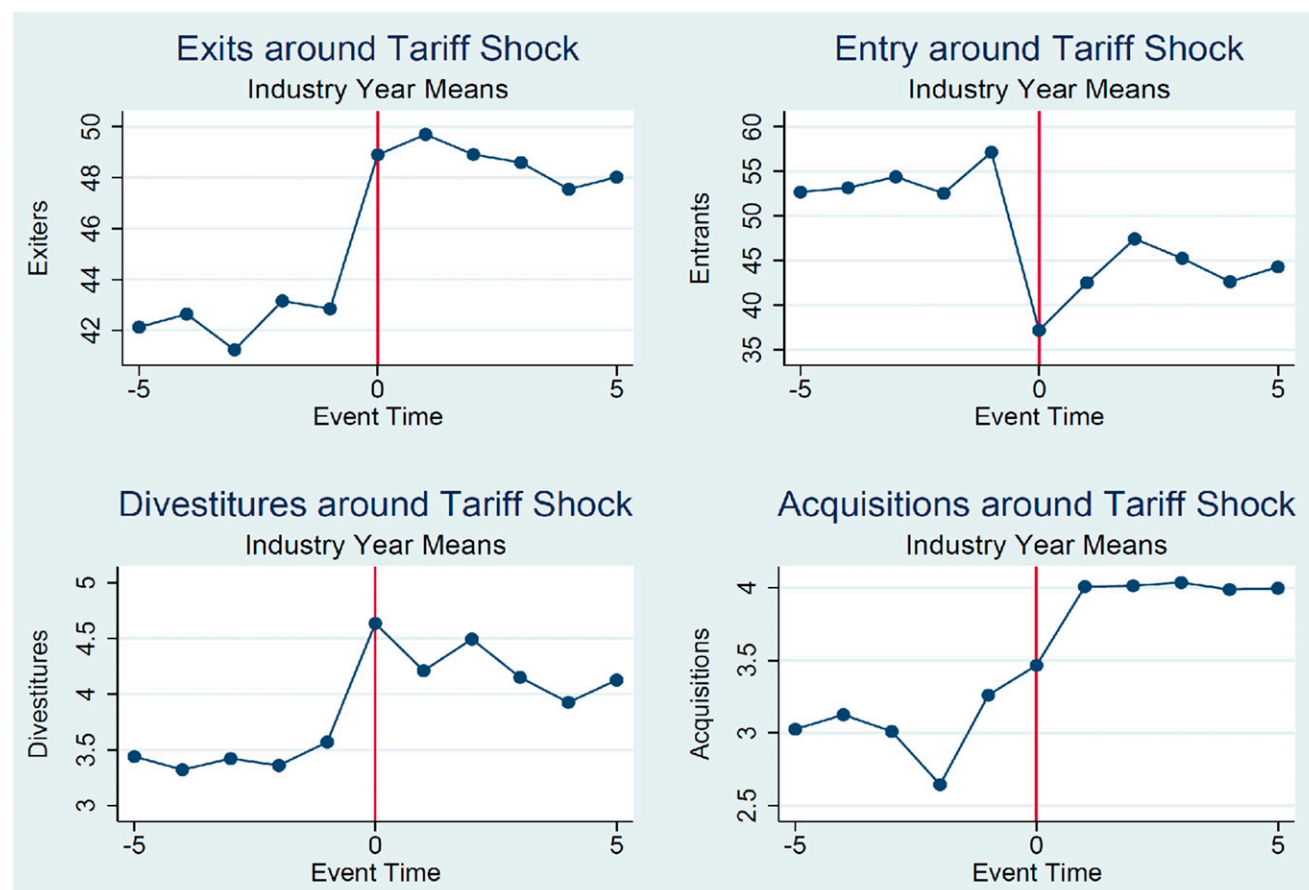
3.2. Data Sources and Sample Construction

3.2.1. Plant-Level Data. Plant-level data comes from the Annual Survey of Manufacturers (ASM) and the Census of Manufacturers (CMF),¹⁵ maintained by the U.S. Census Bureau. The ASM provides comprehensive coverage for the universe of manufacturing plants (SIC codes 2000–3999) in the United States with more than 250 employees and maintains a randomly selected rotating panel for smaller plants.¹⁶ In census years, the ASM is supplanted by the CMF, which surveys the universe of all manufacturing plants in the United States regardless of their size.¹⁷ Combining these two databases allows me to construct a longitudinal panel containing key plant-level variables, such as capital expenditure, material inputs, employment, total assets, value of shipments, and industry. Reporting is mandatory, and companies receive a nontrivial monetary penalty if they misreport.

Because ASM and CMF are constructed as rotating panels and, thus, do not follow the entire life cycle of small plants, I also use the Longitudinal Business Database (LBD) to identify cases of plant closure, opening, and exchanges of ownership (i.e., acquisition and divestiture). The LBD is compiled from the business register and provides annual coverage on employment and payroll information for the universe of all business establishments with at least one paid employee in the United States. Both ASM/CMF and LBD contain unique and permanent plant-level identifiers (i.e., LBDNUM) that allow researchers to merge the two databases.

3.2.2. Import Tariff Data. Data on import tariffs is available from 1972 to 2004 at the 10-digit Harmonized System code level.¹⁸ The data are compiled by Feenstra (1996), Feenstra et al. (2002), and Schott (2010) and are

Figure 2. (Color online) Restructuring Around Tariff Shocks



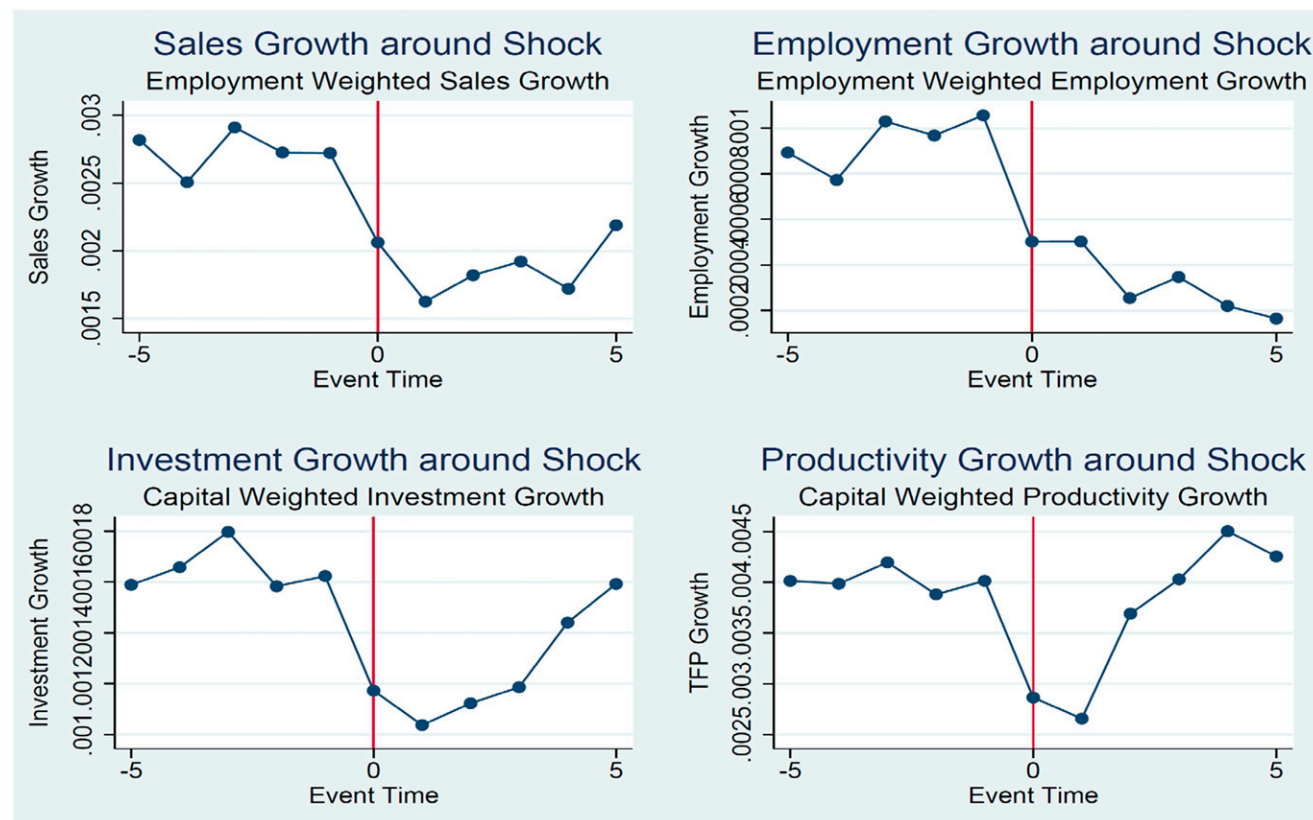
publicly available from both the National Bureau of Economic Research and the Center for International Data at University of California, Davis. To construct import tariff rates at the four-digit SIC industry, I use the concordance table developed in Feenstra et al. (2002) to map the product-level data on imports and duties into 1987 four-digit SIC categories. I then calculate, for each four-digit SIC industry-year, the ad valorem tariff as the duties collected divided by the free-on-board¹⁹ value of imports. This procedure yields a longitudinal panel of import tariff rates (in percentage terms) at the industry-year level.

3.2.3. Main Analytical Sample and Summary Statistics. To construct the analytical sample, I start with the universe of the merged ASM and CMF databases from 1976 to 2004.²⁰ From this initial sample, I exclude observations if the plant operates in an industry-year for which no import tariff data can be obtained, if the plant has a four-digit SIC code ending in nine, if firms have positive exports, or if the data for the plant is imputed from administrative records. I also drop observations for which no productivity (TFP) can be estimated. Finally, plants with fewer than two years of continuous data are dropped to minimize

sampling error.²¹ Because the sampling methodology of ASM makes it difficult to accurately identify entries, exits, and changes of ownership, particularly for small plants, I use the LBD to identify plant status. I also restrict the sample to firms for which plants in the ASM/CMF comprise at least 70% of the firm's total employment.²² I merge this longitudinal panel constructed from the census data to the import tariff data using four-digit SIC industry codes and year. To correct for the oversampling (under-sampling) of large (small) plants associated with the ASM/CMF, all results in this paper are weighted regressions, in which the weights are computed as the reciprocal of the probability weights.²³

Applying these selection criteria yields a final sample of 871,000 establishment-year observations.²⁴ Table 2 presents means and standard deviations for both the full sample and the matched sample described previously in panels A and B, respectively.²⁵

In panel A, column (1) reports summary statistics for the whole sample. Columns (2) and (3) break the data into subsamples of plant-year observations belonging to conglomerate and stand-alone firms, respectively. All dollar figures are expressed in 1997 dollars (in thousands). It is worth noting that plants

Figure 3. (Color online) Other Industry Outcomes Around Trade Shocks

that are part of a conglomerate firm are, on average, quite different compared with their focused counterparts. They are generally larger and older and have lower rates of entry and exit but higher rates in the case of acquisition and divestiture. This pattern is consistent with Maksimovic and Phillips (2001), who find that conglomerates, compared with focused firms, are more likely to participate in the market for corporate control, particularly in partial asset acquisitions and sales. In column (4), *t*-tests for differences show that, with the exception of TFP, conglomerate and focused plants are significantly different across all these other dimensions. I control for plant age and size in all regression analyses. All results remain qualitatively the same if I control for firm age and size instead.

Panel C reports some additional statistics for the full sample. Conglomerates, on average, have a little more than eight plants per firm and operate across 3.424 three-digit SIC industries. The average number of plants in a core industry is roughly 5.66.

3.2.4. Matched Sample. As discussed in Section 3.2.3, conglomerate plants are, on average, larger than their focused peers. This is particularly problematic for the quantification exercise in which I try to measure the economic importance of restructuring activities

(i.e., extensive margin) relative to the intensive margin effects in the functioning of the internal capital market around trade shocks. To mitigate this problem, I construct a matched sample from the main analytical sample. Specifically, for each year, I match conglomerate plants to stand-alone plants on plant size (i.e., the logarithm of total value of shipments) and industry (i.e., four-digit SIC industry). Summary statistics on key variables are in Table 2, panel B. Conglomerates and stand-alones in the matched sample are similar in size but exhibit differences across other dimensions, such as plant age and restructuring probabilities. Because of this remaining difference, one needs to interpret with caution results using the matched sample.

3.3. Key Variable Definition

In this section, I define the key variables used in the analyses. The appendix contains detailed definitions for all other variables.

3.3.1. Conglomerate Dummy. I classify a firm as a conglomerate if it operates in more than one unique three-digit SIC industry and as a stand-alone firm otherwise. I make this choice for two reasons: First, defining firm organization at the three-digit SIC level is a compromise between too coarse a partition, in which unrelated industries may be pooled together,

Table 2. Summary Statistics

Panel A: Full sample				
	All plants (1)	Conglomerates (2)	Stand-alones (3)	T-test for difference (4)
Value of shipments	49,199 (543,624)	90,287 (429,339)	24,330 (600,992)	20.54
TFP	2.344 (0.7107)	2.315 (0.7319)	2.361 (0.6970)	−1.05
Total employment	215 (605)	377 (897)	117 (275)	9.84
Plant size	8.906 (1.926)	10.029 (1.637)	8.226 (1.763)	6.23
Plant age	8.030 (7.519)	11.564 (8.171)	5.892 (6.182)	32.38
Plant opening	0.089 (0.337)	0.051 (0.221)	0.132 (0.383)	−33.34
Plant closure	0.092 (0.361)	0.064 (0.277)	0.115 (0.397)	−30.07
Plant divestiture	0.029 (0.168)	0.040 (0.197)	0.022 (0.147)	15.53
Plant acquisition	0.028 (0.166)	0.041 (0.199)	0.020 (0.142)	16.03
Observations	871,000	329,000	543,000	
Panel B: Matched sample				
	All plants (1)	Conglomerates (2)	Stand-alones (3)	T-test for difference (4)
Value of shipments	71,820 (260,400)	87,080 (300,200)	39,090 (135,900)	4.41
TFP	2.312 (0.7216)	2.306 (0.7317)	2.325 (0.6993)	−0.83
Total employment	327 (784)	379 (906)	215 (391)	7.25
Plant size	9.780 (1.669)	9.989 (1.658)	9.532 (1.602)	1.58
Plant age	10.15 (7.748)	11.18 (7.834)	7.95 (7.074)	22.39
Plant opening	0.0689 (0.2533)	0.0505 (0.2189)	0.1084 (0.3109)	−26.35
Plant closure	0.1 (0.3000)	0.082 (0.2744)	0.1385 (0.3455)	−21.15
Plant divestiture	0.0386 (0.1928)	0.0414 (0.1992)	0.0327 (0.1780)	7.79
Plant acquisition	0.0382 (0.1916)	0.0411 (0.1986)	0.0318 (0.1754)	7.01
Observations	445,000	304,000	142,000	
Panel C: Firm composition				
	Full sample (1)	Conglomerates (2)	Stand-alones (3)	
Unique firms	113,000	3,000	110,000	
Unique firm-year observations	520,000	40,000	480,000	

Table 2. (Continued)

Panel C: Firm composition			
	Full sample (1)	Conglomerates (2)	Stand-alones (3)
Number of plants per firm	1.675	8.256	1.130
Number of three-digit SIC industries		3.424	
Number of plants in core industry		6.120	

Notes. This table lists the sample mean and standard deviation (in parentheses) of the main variables in the analyses. The sample period is from 1976 to 2004. Value of shipments is total plant level sales deflated using the four-digit SIC industry shipment deflators from the NBER-CES Manufacturing Industry Database expressed in 1997 dollars. Employment is the total number of people employed at each plant. Plant size is the logarithm of the total value of shipments of the plant. Plant age is the number of years since a plant's first appearance in the LBD. Opening dummy is a dummy variable that equals one if a plant is in its first year of existence. Closure dummy is a dummy variable that equals one if a plant is in its last year of existence. Divestiture dummy is a dummy variable if a plant is sold to another firm in year t , and zero otherwise. Acquisition dummy is a dummy variable that equals one if a plant is in its first year of existence for firm i in year t but belonged to a different firm in year $t - 1$. Conglomerates are plants that belong to a firm that operates in more than one distinct three-digit SIC industry in a given year. Stand-alones are plants that belong to a firm that operates in one distinct three-digit SIC industry in a given year.

and too narrow a partition, which may be subject to misclassification. Second, this definition is consistent with the approach used in prior studies using census data (e.g., Maksimovic and Phillips 2001, 2002), which makes the findings of the paper more comparable and easier to interpret in light of the prior studies. Note that all results remain unchanged if I define the conglomerate dummy based on the two-digit SIC as in Schoar (2002). In robustness checks, I reestimate the main tests by defining conglomerates at the four-digit SIC industry level and also find similar results.

3.3.2. Measure of Efficiency: Total Factor Productivity. The primary measure of efficiency used in the analyses is TFP estimated with a log-linear Cobb–Douglas specification (see Schoar 2002, Li 2013, Giroud and Mueller 2015). Specifically, TFP of a given plant i in year t is the estimated residual from the following regression:

$$y_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \beta_m m_{it} + \epsilon_{it}, \quad (1)$$

where i and t index plant and year, respectively, and y , k , l , and m are logarithms of output, capital, labor, and material inputs, respectively. I estimate Equation (1) for each three-digit SIC industry and year. This allows for different factor intensities across industries and over time, which is particularly important given my setting.²⁶ I acknowledge the potential limitation associated with imposing a structure on the production function as in Equation (1). For instance, productivity estimation based on Equation (1) could be subject to the simultaneity bias and the selection bias (Akerberg et al. 2007). In robustness checks, I employ the structural techniques of Olley

and Pakes (1996), which address the simultaneity problem by using investment to proxy for the productivity shock.²⁷ The selection problem is addressed by estimating plant survival propensity scores. I find that the main effects are robust to using the structural approach of productivity estimation.

3.3.3. Treated Dummy. A plant is defined as treated if it operates in a four-digit SIC industry that experiences a trade shock in any of the past three years (t , $t - 1$, or $t - 2$). Accordingly, a firm is a treated firm if one or more of its plants experiences a trade shock within the past three years. The rationale for this definition is to focus on the short-term operation of the internal capital market in response to the trade shock. Although looking at longer horizons has advantages, it also introduces more noise into the inference and interpretation.

3.3.4. Core vs. Noncore. For conglomerate firms, I distinguish between core industries from the peripheral operations. Specifically, within a firm, a three-digit SIC industry is classified as a core industry if it accounts for 50% or more of the firm's total output. Otherwise, the industry is classified as a peripheral industry. Accordingly, a plant is defined as a core plant (*Core*) if it operates in a core industry and as a peripheral plant (*Noncore*) if it operates in a peripheral industry. In untabulated results, I experiment with other thresholds (i.e., 20%, 30%, and up to 40%) and find qualitatively similar results.

3.3.5. Plant Opening, Closure, Acquisition, and Divestiture. As discussed previously, I use the LBD to define plant statuses. Specifically, plant opening is defined as an indicator variable that equals one in a plant's first year

of existence. Symmetrically, I define plant closure as occurring if a plant is in its last year of existence. I separate plant opening from the purchase of a plant (i.e., acquisition). The LBD includes unique firm- and plant-level identifiers, allowing one to trace the same plant under the ownership of different firms. I also distinguish between the permanent closure of a plant and the sale of a plant to another firm. Differentiating between new entry and acquisition of existing plants is important for two reasons: First, conglomerate firms are more likely to engage in acquisitions relative to stand-alones (e.g., Maksimovic and Phillips 2001, 2008). Second, it is crucial to separate these cases in the quantification exercise of conglomerates' internal capital market.

4. Empirical Specification and Results

In order to get a sense of the impact of trade shocks on the average productivity of plants in the affected industry and how it differs across firms of distinct organizational forms, I employ a nonparametric approach and graph the average plant-level productivity surrounding these events.²⁸ Figure 4 shows that the average TFP of conglomerate and stand-alone plants follows a similar path prior to the tariff shock, suggesting that the parallel trend assumption likely holds in this setting. This pattern also confirms prior studies suggesting that the import tariff shocks cannot be fully anticipated, and no pretrend exists leading up to the trade shocks. Interestingly, although focused plants' TFP drops sharply around the shock and continues to decline during the following five-year window after that, conglomerate plants' TFP experiences a mild drop at the onset of the shock but quickly rebounds to a slightly higher level. This could arise for either or both of the following reasons: First, it is possible that the continuing plants of conglomerate firms become more efficient than focused firms by, for example, hiring more capable managers or adopting more advanced production technology. Alternatively, conglomerates could engage in

dynamic restructuring, such as opening more efficient plants and closing less efficient ones to minimize the impact of the import tariff shock on the affected sector. Remaining tests in this section attempt to uncover the underlying causes by estimating multivariate fixed effects regressions.

4.1. Effect of Trade Shock on Continuing Plants: The Intensive Margin

In the first set of tests, I adopt a triple-difference research design to examine how productivity of conglomerates and focused firms is differentially affected by the import tariff shocks. To this end, I estimate the following panel regression model:

$$y_{ijkst} = \alpha_i + \alpha_{st} + \beta_1 \times Treated_{ikt} + \beta_2 \times Conglomerate_{i,j,t-1} + \beta_3 \times (Conglomerate_{i,j,t-1} \times Treated_{ikt}) + \gamma' X_{ijkst} + \epsilon_{ijkst} \quad (2)$$

where i, j, k, s , and t index plant, firm, industry, state, and time, respectively; i is plant fixed effects; α_{st} is state-year fixed effects; X_{ijkst} are control variables (i.e., plant age and size); $Treated_{ikt}$ is an indicator variable that equals one if a plant operates in an industry that experiences a trade shock within the past three years ($t, t-1$, and $t-2$) and zero otherwise; ϵ_{ijkst} is the error term.

Because the purpose of this test is to examine the impact of trade shocks on the continuing plants' productivity, I include plant fixed effects to control for time-invariant plant-specific characteristics, such as unobservable managerial ability. I also include state-year fixed effects to account for contemporaneous local shocks (e.g., local productivity shocks) at the state level. In principle, I could control for a full set of industry $\times$ state $\times$ year fixed effects; however, doing so would require the inclusion of 492 four-digit SIC $\times$ 29 years $\times$ 50 states = 713,400 additional fixed effects. This is infeasible given the sample size and the limited computing power at the Federal Statistical Research Data Center.²⁹ As a compromise, I include as one of the control variables the industry-state-year average of the dependent variable in all regressions.³⁰ In Equation (2), the total effect of the trade shocks on productivity is measured by $\beta_1 + \beta_3 \times Conglomerate$. The coefficient β_1 measures the effect of the import tariff reduction on plant efficiency for focused plants. The coefficient β_2 captures the inherent differences in productivity between conglomerate and focused plants.³¹ Finally, the coefficient β_3 measures the differential change in productivity of conglomerate plants and focused plants both affected by the trade shock. I cluster standard errors at the four-digit SIC level. Note that clustering at the three-digit SIC level, firm level, or industry-state level does not change the results.

Figure 4. (Color online) Productivity of Treated Plants Around Trade Shocks

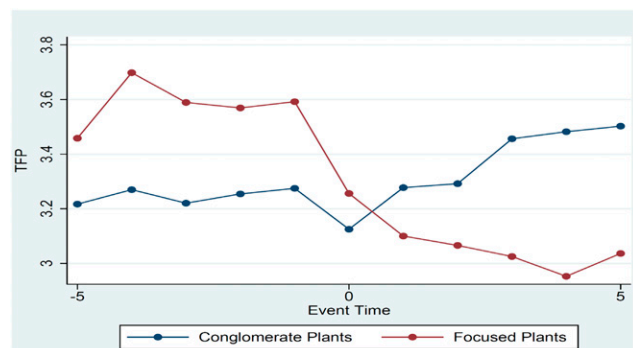


Table 3, column (1) estimates the effect of the trade shock on the TFP of treated plants using the full sample, and columns (2) and (3) estimate on subsamples of conglomerate plants and stand-alone plants, respectively. In column (1), the inclusion of the plant fixed effects implies that the identification comes from plants that continue to operate during the trade shock. Note the coefficient on *Conglomerate* is negative and significant at 5%, implying that conglomerate plants absent the shock are slightly less productive on average.³² The large and negative coefficient on *Treated* suggests that focused plants (i.e., *Conglomerate* = 0) experience a significant decline in productivity. Finally, *Treated* × *Conglomerate* is positive and statistically significant at the 1% level, which suggests that conglomerate plants can somehow weather the impact of the shock. The economic magnitude is also nontrivial: treated conglomerate plants experience, on average, a productivity drop that is 2.6 log points smaller than their focused peers. To put these numbers in perspective, the average TFP is 2.334 with a standard deviation of 0.71, so the preceding estimate represents a sizeable difference in TFP. Columns (2) and (3) estimate the impact of the tariff shock on productivity of conglomerates and focused plants, respectively. Productivity of treated conglomerate plants experience a small decline (−0.006 log points),

whereas that of treated stand-alone plants decreases significantly (−0.052 log points). Next, I reestimate Equation (2) using the subsample of continuing plants and a sample of conglomerates and stand-alones matched on industry and plant size (see Section 3.2.4.) in columns (4) and (5), respectively. In both cases, conglomerate plants affected by the import tariff shock experience a smaller decrease in TFP compared with their focused counterparts. Taken together, using various specifications and subsamples, I find consistent results suggesting that conglomerates can somehow shield the negative impact of the tariff shock on productivity of the treated plants, which is broadly consistent with Hypothesis 1 based on the predictions of the winner-picking hypothesis.

4.2. Corporate Restructuring: The Extensive Margin

Although the previous section shows that conglomerate plants affected by trade shocks that continue to operate (i.e., continuers) experience a smaller decrease in productivity compared with their focused peers, I now focus on firms' restructuring activities. As discussed in Section 2.2.2, conglomerates' response to the import tariff shocks can be through either the intensive or extensive margin. Therefore, in this section, I concentrate on plant closure, opening, divestiture, or acquisition. Relative to intensive margin activities,

Table 3. Productivity of the Treated Plants

Dependent variable	TFP (Cobb–Douglas)				
	Full sample (1)	Conglomerates (2)	Stand-alones (3)	Continuers only (4)	Matched sample (5)
<i>Treated</i>	−0.033*** (0.015)	−0.006 (0.004)	−0.052*** (0.012)	−0.034*** (0.006)	−0.034** (0.015)
<i>Treated</i> × <i>Conglomerate</i> _{<i>t</i>−1}	0.026*** (0.009)			0.024*** (0.006)	0.058** (0.017)
<i>Conglomerate</i> _{<i>t</i>−1}	−0.016** (0.006)			−0.0151*** (0.003)	−0.011*** (0.027)
<i>Plant Size</i> and <i>Plant Age</i> control	Yes	Yes	Yes	Yes	Yes
Industry-state-year mean	Yes	Yes	Yes	Yes	Yes
Plant fixed effects	Yes	Yes	Yes	Yes	Yes
State × Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	871,000	329,000	543,000	192,000	445,000
Adjusted R ²	0.27	0.25	0.33	0.46	0.37

Notes. This table presents plant-level difference-in-difference regressions as outlined in Equation (2). The dependent variable is TFP of the plant estimated using the Cobb–Douglas production function described in Equation (1). *Treated* is an indicator variable that equals one if a given plant experiences a trade shock in the form of major import tariff reduction in the past three years (*t*, *t* − 1, or *t* − 2) and zero otherwise. *Conglomerate* is an indicator variable that takes the value of one if a firm operates in more than one three-digit SIC industry in a given year and is zero otherwise. The control variables are plant size and plant age. *Plant size* is the natural logarithm of the plant's value of shipments. *Plant age* is the natural logarithm of one plus the number of years since the plant has been in the LBD. Both variables are lagged. In column (4), continuers are plants that exist at the onset of the tariff shock and survive until two years after the shock (i.e., from *t* to *t* + 2). Industry-state-year mean is the average value of the dependent variable in a four-digit SIC-state-year. All other variables are defined in the appendix. Heteroscedasticity robust standard errors are in parentheses and are corrected for clustering at the four-digit SIC industry level.

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

such as laying off workers and cutting back on capital expenditure, these extensive margin adjustments are some of the most radical real actions that firms can take in response to tariff shocks. I investigate the likelihood of corporate restructuring and the role that segment importance plays in conglomerates' restructuring in Sections 4.2.1 and 4.2.2, respectively. Together, these tests help differentiate Hypothesis 2 and Hypothesis 2a.

4.2.1. Probability of Restructuring. The first step of my analysis focuses on how conglomerates' and focused firms' propensity to restructure is affected by the tariff shock. Specifically, I estimate the following panel regression model:

$$y_{ijkst} = \alpha_k + \alpha_{st} + \beta_1 \times Treated_{ikt} + \beta_2 \times Conglomerate_{ij,t-1} + \beta_3 \times (Conglomerate_{ij,t-1} \times Treated_{ikt}) + \gamma' X_{ijkst} + \epsilon_{ijkst}, \quad (3)$$

where i, j, k, s , and t index plant, firm, industry, state, and time, respectively; α_k is industry fixed effects to account for time-invariant industry-specific characteristics; α_{st} is state-year fixed effects and control for time-varying local economic conditions; X_{ijkst} is control variables (i.e., plant age and size); $Treated_{ikt}$ is an indicator variable that equals one if a plant operates in an industry that experiences a trade shock within the past three years ($t, t-1$, and $t-2$); ϵ_{ijkst} is the error term. Although linear probability models have their own advantages and disadvantages compared with probit and tobit models,³³ they do provide a more transparent interpretation in coefficient estimates (directly as probabilities, similar to the marginal effects in the case of probit) and perform better when a number of fixed effects are included in the estimation. The dependent variables are indicator variables of plant opening, closure, acquisition, or divestiture. These variables are defined in more detail in Section 3.3.5.³⁴

Table 4 contains the results of this exercise. Panel A investigates plant closure and divestiture, and panel B focuses on plant opening and acquisition. Several results are worth noting in panel A. First, the import tariff reductions significantly change the propensity to restructure the affected plants for both types of firms; the likelihood of closing a treated plant increases by 7.5% (column (2)) and 3.6% (column (3)) in probability for conglomerates and stand-alones, respectively. These effects are economically large, representing approximately 117% (0.075/0.064) and 31% (0.036/0.115) of their respective baseline probability of plant closure. Second, the difference in the change of probability of plant closure between conglomerates and focused firms is positive and significant as indicated by the coefficient of 0.024 on the interaction term in column (1)

($Conglomerate \times Treated$). Conglomerate firms are unconditionally (without the tariff shock) less likely to shut down plants (−0.040 on *Conglomerate*) but are significantly more likely to close treated plants affected by the trade shock. Columns (4)–(6) paint a similar picture for plant divestiture. Overall, in terms of plant closure and divestiture, it appears that conglomerates reduce their presence in the affected industries by shutting down plants affected by the import tariff shock, consistent with the winner-picking hypothesis.

Panel B of Table 4 concentrates on episodes of plant opening and acquisition. Conceptually, this analysis informs us about how the willingness to build (acquire) new capacity in the affected industry changes in expectation of the intensifying competition. In particular, given that conglomerates are more likely to close down and divest assets in the affected industry, one would expect them to exhibit a lower probability of opening new plants or acquiring new plants in the treated sectors. These results would together be fully consistent with the winner-picking hypothesis. The results as shown in panel B of Table 4 tell a more nuanced story: conglomerate firms have, on average, a lower propensity to open new plants (−0.017 on *Conglomerate* in column (1)). The likelihood of opening plants in the treated industry experiences only a small decline (−0.004 in column (2)),³⁵ whereas focused firms' probability of opening new plants drops by more than 8% (column (3)). In terms of economic magnitude, conglomerates' decline in probability of opening new plants in the affected industry corresponds to only 7.8% of their baseline probability of new entry (0.004/0.051), whereas the same metric for focused firms is 73% (0.084/0.132). Plant acquisitions exhibit a similar pattern: conglomerates are more likely relative to focused firms to acquire plants in the affected industry. These results together suggest that conglomerate firms seem to be more responsive than their focused counterparts in restructuring activities. In particular, by exploiting their more flexible organizational structure, conglomerate firms reshuffle their operational assets more aggressively.

4.2.2. Conglomerates' Restructuring: Core vs. Non-Core.

Although the results in Section 4.2.1 suggest conglomerates' relatively greater responsiveness in adjusting through the extensive margin, they mask the important heterogeneity in segment importance within a conglomerate firm. In particular, as discussed in Section 2.2.2, conglomerates' allocation of resources might favor their core segments with a goal of protecting their corporate reputation and long-term product market success. In this section, I focus on conglomerate firms and test whether this is true in the data. Specifically, I estimate, following

Table 4. Likelihood of Restructuring

Panel A: Probability of plant closure and divestiture						
Dependent variable	Closure Dummy			Divestiture Dummy		
	Full sample (1)	Conglomerates (2)	Stand-alones (3)	Full sample (4)	Conglomerates (5)	Stand-alones (6)
<i>Treated</i>	0.068*** (0.016)	0.075*** (0.018)	0.036*** (0.016)	0.004 (0.003)	0.011*** (0.001)	0.004 (0.003)
<i>Treated</i> × <i>Conglomerate</i> _{<i>t</i>−1}	0.024** (0.006)			0.014** (0.007)		
<i>Conglomerate</i> _{<i>t</i>−1}	−0.040*** (0.004)			0.011*** (0.001)		
<i>Plant Size</i> and <i>Plant Age</i> control	Yes	Yes	Yes	Yes	Yes	Yes
Industry-state-year mean	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State × Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	871,000	329,000	543,000	871,000	329,000	543,000
Adjusted <i>R</i> ²	0.36	0.29	0.38	0.06	0.08	0.05
Panel B: Probability of opening and acquisition						
Dependent variable	Opening Dummy			Acquisition Dummy		
	Full sample (1)	Conglomerates (2)	Stand-alones (3)	Full sample (4)	Conglomerates (5)	Stand-alones (6)
<i>Treated Dummy</i>	−0.086*** (0.028)	−0.004* (0.002)	−0.084*** (0.026)	0.000 (0.002)	0.008*** (0.002)	0.003 (0.002)
<i>Treated</i> × <i>Conglomerate</i> _{<i>t</i>−1}	0.081*** (0.022)			0.008** (0.003)		
<i>Conglomerate Dummy</i> _{<i>t</i>−1}	−0.017*** (0.003)			0.021*** (0.002)		
<i>Plant Size</i> and <i>Plant Age</i> control	Yes	Yes	Yes	Yes	Yes	Yes
Industry-state-year mean	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State × Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	871,000	329,000	543,000	871,000	329,000	543,000
Adjusted <i>R</i> ²	0.09	0.10	0.08	0.03	0.01	0.04

Notes. This table presents plant-level difference-in-difference regressions as outlined in Equation (3). The dependent variables are the closure dummy, divestiture dummy, opening dummy, or acquisition dummy. *Conglomerate* is an indicator variable that takes a value of one if a firm operates in more than one three-digit SIC industry. *Closure Dummy* is a dummy variable if a plant is in its last year of existence in the LBD. *Divestiture Dummy* is a dummy variable if a plant is sold to another firm in year *t* and zero otherwise. *Opening Dummy* is a dummy variable if a plant is in its first year of existence in the LBD. *Acquisition Dummy* is a dummy variable that equals one if a plant is in its first year of existence for firm *i* in year *t* but belonged to a different firm in year *t* − 1. *Treated* is an indicator variable that equals one if a given plant experiences a trade shock in the form of major import tariff reduction in the past three years (*t*, *t* − 1, or *t* − 2) and zero otherwise. Industry-state-year mean is the average value of the dependent variable in a four-digit SIC industry-state-year. Both variables are lagged. Heteroscedasticity robust standard errors are in parentheses and are corrected for clustering at the four-digit SIC industry level.

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Bertrand and Mullainathan (2003), the following ordinary least squares panel regression model:

$$\begin{aligned}
 y_{ijkst} = & \alpha_j + \alpha_{st} + \beta_0 \times \text{Core}_{ij,t-1} + \beta_1 \times (\text{Treated}_{ikt} \times \text{Core}_{ij,t-1}) \\
 & + \beta_2 \times (\text{Treated}_{ikt} \times \text{Noncore}_{ij,t-1}) \\
 & + \beta_3 \times (\text{Other}_{ikt} \times \text{Core}_{ij,t-1}) \\
 & + \beta_4 \times (\text{Other}_{ikt} \times \text{Noncore}_{ij,t-1}) + \gamma' X_{ijkst} + \epsilon_{ijkst}, \quad (4)
 \end{aligned}$$

where *i*, *j*, *k*, and *s* index plant, firm, industry, and state, respectively; α_j and α_{st} are firm and state-year fixed effects, respectively. Note that I include firm

fixed effects to control for time-invariant firm characteristics. For instance, certain firms are inherently more likely to restructure because of their corporate policy or managerial preference. In addition, the fixed effects ensure that the comparison is between plants within the same firm. The dependent variable y_{ijkst} is an indicator of plant exit or entry; Treated_{ikt} are defined the same as in Equation (2); $\text{Core}_{ij,t-1}$ is an indicator variable that takes a value of one if a plant operates in a three-digit SIC industry that is the largest within the firm in terms of sales and also contributes 50% or more of the firm's total sales; $\text{Noncore}_{ij,t-1}$ is the

complement of $Core_{ijt-1}$ (i.e., $1 - Core_{ijt-1}$); $Other_{ikt}$ is an indicator variable that equals one if a plant is unaffected by the trade shock itself but belongs to a firm that has one or more treated plants and is zero otherwise; β_1 and β_2 identify the direct effect of the tariff shock on a plant operating in the core and peripheral industry, respectively. One innovation of the empirical specification in Equation (4) is the inclusion of $Other_{ikt}$ interacted with $Core$ and $Noncore$. This allows one to examine whether conglomerates' likely incentive to focus on their core business generates any spillover effect for their peripheral operations. For instance, β_4 would measure the impact of the trade shock on a peripheral plant when one or more other industries are hit by a tariff shock.

If conglomerate firms tend to protect their core business operations, I conjecture that they are less (more) likely to shut down (open up) plants in the core industry even when the industry experiences a negative trade shock. In contrast, conglomerates should treat their peripheral plants in a way that is more consistent with the winner-picking hypothesis. Table 5 presents the results of this exercise. Panel A focuses on plant closure and divestiture. As shown in column (1), not only do plants affected by the trade shocks experience a large increase (0.068) in the probability of being closed down, plants that are unaffected but are part of the conglomerate umbrella also are more likely to be closed (0.059). Both changes in probability are economically sizable and statistically significant, representing roughly 106% and 92% of the average probability of closure of conglomerate plants, which suggests that trade shocks have both a direct and an indirect effect on plants within a conglomerate. In column (2), I decompose these effects by including $Core$ and $Noncore$ and estimate the specification in Equation (4). The results confirm conglomerates' strong tendency to refocus on their core operations: in terms of the direct effect, although treated plants in core industries ($Treated \times Core$) experience only a change in the probability of plant closure indistinguishable from zero (0.023), noncore assets affected by the shock ($Treated \times Noncore$) are significantly more likely to be shut down (0.074). This differential change in probability between core and noncore plants reflects conglomerates' desire to stay in their core business in response to the competitive pressure. Perhaps more interesting, a strong indirect effect exists: peripheral plants that are themselves unaffected but belong to a conglomerate firm that has some other plants affected by the trade shock ($Other \times Noncore$) experience a 8.1% increase in the probability of closure. On the other hand, core plants exhibit no such spillover effect ($Other \times Core$). Columns (3) and (4) examine plant divestiture and show a similar pattern but with smaller magnitude.

Panel B of Table 5 concentrates on episodes of plant opening and acquisition and uncovers several striking findings. First, similar to the case of plant closure, the tariff cuts have a spillover effect on the propensity of conglomerates to open new plants in other industries that do not experience the shocks (-0.056 on $Other$ in column (1)). Second, conglomerates' plant opening and acquisition decisions also reflect their incentive to refocus on the core business. In terms of direct effect, conglomerates are 11.5% less likely to open new establishments in their peripheral industries that are affected by the trade shock ($Treated \times Noncore$) but are 9.1% more likely to open new plants in treated core industries ($Treated \times Core$). In terms of the indirect effect, peripheral businesses experience a significant negative spillover (-0.170 on $Other \times Noncore$), but no such effect is observed for core plants (the insignificant 0.031 on $Other \times Core$). The same tests for plant acquisitions show a much weaker result. One possibility is that adjustment through plant acquisitions is more time-consuming and depends on other economic factors (e.g., finding a counter party and settling on a mutually agreeable price). Hence, the short event window on which I focus may not be enough to discern such corporate actions.

Taken together, the results in Table 5 show that conglomerate firms exhibit a strong tendency to readjust their firm scope toward their core business when affected by the tariff shock. In particular, conglomerates are more likely to reduce their footprint in peripheral industries even when these industries themselves are not affected by the tariff shocks. Meanwhile, they tend to increase their presence in the treated core industries despite the presumably worse prospects in the short run as the strategic value of protecting the core operations could be of greater significance. Although these results are consistent with Hypothesis 2a, they do not offer a conclusion as to whether these actions are efficiency-improving or efficiency-destroying. The next section sheds light on these issues by investigating whether conglomerates' restructuring decisions depend on plants' TFP as well as industry prospects.

4.2.3. Conglomerates' Restructuring: Within-Firm Winner-Picking or Agency Issues. In this section, I investigate whether conglomerates' restructuring activities and their tendency to build more capacity in their core segment reflect their strategic vision to focus on long-run success or are simple manifestations of their presumably higher agency issues. I conduct three sets of tests: First, I compare conglomerates' restructuring decisions to those of multiplant-focused firms to examine whether conglomerates' adjustments are more driven by establishment productivity. Second, I test whether, even within their

Table 5. Conglomerates' Restructuring: Core vs. Noncore

Panel A: Plant closure and divestiture: Core versus noncore				
Dependent variable	Closure Dummy		Divestiture Dummy	
	(1)	(2)	(3)	(4)
<i>Treated</i>	0.068*** (0.017)		0.008*** (0.001)	
<i>Other</i>	0.059*** (0.019)		0.012*** (0.001)	
<i>Treated</i> × <i>Core</i> _{<i>t</i>−1}		0.023 (0.016)		0.003 (0.002)
<i>Treated</i> × <i>Noncore</i> _{<i>t</i>−1}		0.074*** (0.020)		0.006*** (0.001)
<i>Other</i> × <i>Core</i> _{<i>t</i>−1}		−0.006 (0.022)		−0.000 (0.001)
<i>Other</i> × <i>Noncore</i> _{<i>t</i>−1}		0.081*** (0.016)		0.013*** (0.002)
Size and age control	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
State-year fixed effects	Yes	Yes	Yes	Yes
Observations	329,000	329,000	329,000	329,000
Adjusted <i>R</i> ²	0.28	0.26	0.08	0.08
Panel B: Plant opening and acquisition: Core versus noncore				
Dependent variable	Opening Dummy		Acquisition Dummy	
	(1)	(2)	(3)	(4)
<i>Treated</i>	−0.032 (0.023)		0.002 (0.002)	
<i>Other</i>	−0.056*** (0.021)		0.000 (0.003)	
<i>Treated</i> × <i>Core</i> _{<i>t</i>−1}		0.091*** (0.020)		0.002* (0.001)
<i>Treated</i> × <i>Noncore</i> _{<i>t</i>−1}		−0.115*** (0.028)		−0.001 (0.003)
<i>Other</i> × <i>Core</i> _{<i>t</i>−1}		0.031 (0.017)		0.001 (0.002)
<i>Other</i> × <i>Noncore</i> _{<i>t</i>−1}		−0.170*** (0.032)		−0.001 (0.005)
Size and age control	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
State-year fixed effects	Yes	Yes	Yes	Yes
Observations	329,000	329,000	329,000	329,000
Adjusted <i>R</i> ²	0.10	0.10	0.03	0.02

Notes. This table presents plant-level difference-in-difference regressions as outlined in Equation (4). The dependent variables are the closure dummy, divestiture dummy, opening dummy, or acquisition dummy. Panel A examines cases of plant closure and divestiture, and panel B focuses on plant opening and acquisition. *Core* is an indicator variable that equals one if a plant operates in a three-digit SIC industry that accounts for 50% or more of the total firm sale. *Noncore* is the complement of *Core* (i.e., $1 - \text{Core}$). *Treated* is an indicator variable that equals one if a given plant experiences a trade shock in the form of major import tariff reduction in the past three years ($t, t - 1$, or $t - 2$) and zero otherwise. *Other* is an indicator variable that equals one if a plant does not experience a trade shock in the past three years itself but belongs to a firm that has some other plants experiencing the trade shock. All regressions include plant size and plant age as control variables. All other variables are defined in the appendix. Heteroscedasticity robust standard errors are in parentheses and are corrected for clustering at the four-digit SIC industry level.

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

core operations, conglomerates' restructuring decisions are driven by plant productivity. Finally, I investigate whether conglomerates' restructuring is potentially forward-looking in that they tend to build

more capacity in industries that have a high growth potential in the future.

The results of these exercises are in Table 6. In panel A, I study whether conglomerates' restructuring

Table 6. Conglomerates' Restructuring: Within-Firm Winner-Picking

Panel A: Effect of plant TFP								
	Conglomerates		Multiplant-focused firms		Conglomerates		Multiplant-focused firms	
	Closure (1)	Divestiture (2)	Closure (3)	Divestiture (4)	Opening (5)	Acquisition (6)	Opening (7)	Acquisition (8)
<i>Treated</i> × <i>Low TFP Tercile</i> _{<i>t</i>−1}	0.137*** (0.019)	0.018 (0.012)	0.042* (0.022)	0.005 (0.007)	−0.104*** (0.023)	−0.002 (0.003)	−0.184*** (0.025)	−0.005* (0.002)
<i>Treated</i> × <i>Medium TFP Tercile</i> _{<i>t</i>−1}	0.005 (0.022)	0.002 (0.010)	0.037* (0.026)	0.006 (0.019)	0.033 (0.020)	0.005 (0.003)	−0.098*** (0.021)	0.004 (0.003)
<i>Treated</i> × <i>High TFP Tercile</i> _{<i>t</i>−1}	−0.031*** (0.011)	−0.005 (0.003)	0.036* (0.019)	0.017 (0.013)	0.045*** (0.016)	0.003* (0.002)	−0.052*** (0.021)	0.003 (0.002)
<i>Medium TFP Tercile</i> _{<i>t</i>−1}	−0.010 (0.007)	0.005 (0.003)	−0.003 (0.016)	0.004 (0.014)	0.026 (0.015)	0.005 (0.003)	0.012 (0.019)	0.015 (0.018)
<i>High TFP Tercile</i> _{<i>t</i>−1}	−0.067*** (0.007)	−0.022** (0.005)	−0.054** (0.016)	−0.026** (0.013)	−0.023 (0.019)	0.002 (0.004)	0.037* (0.020)	0.029* (0.015)
Size and age control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	329,000	329,000	94,000	94,000	329,000	329,000	94,000	94,000
Adjusted R ²	0.28	0.06	0.18	0.04	0.15	0.05	0.13	0.05
Panel B: Conglomerates: Winner-picking among core segments								
Dependent variable	Closure		Divestiture		Opening		Acquisition	
<i>Core</i>	−0.013*** (0.002)		−0.011*** (0.002)		0.072*** (0.020)		0.002 (0.002)	
<i>Above Median TFP</i>	−0.045*** (0.016)		0.002** (0.001)		0.022 (0.019)		0.025** (0.012)	
<i>Treated</i>	0.040*** (0.012)		0.007* (0.004)		−0.003 (0.003)		0.003 (0.002)	
<i>Core</i> × <i>Treated</i>	0.009 (0.006)		−0.002 (0.007)		0.061*** (0.023)		0.003 (0.003)	
<i>Core</i> × <i>Above Median TFP</i>	−0.030** (0.015)		−0.003** (0.002)		0.031* (0.015)		0.004 (0.002)	
<i>Treated</i> × <i>Above Median TFP</i>	−0.028** (0.013)		0.001 (0.003)		0.025** (0.009)		0.005* (0.003)	
<i>Above Median TFP</i> × <i>Core</i> × <i>Treated</i>	−0.026** (0.011)		0.002 (0.004)		0.040* (0.024)		0.001 (0.005)	
Size and age control	Yes		Yes		Yes		Yes	
Firm fixed effects	Yes		Yes		Yes		Yes	
State-year fixed effects	Yes		Yes		Yes		Yes	
Observations	329,000		329,000		329,000		329,000	
Adjusted R ²	0.30		0.09		0.13		0.06	
Panel C: Conglomerates: Industry prospects								
Dependent variable	Exit (1)		Entry (2)		Purchase (3)		Sale (4)	
<i>Treated</i> × <i>Growth Industry</i>	−0.052** (0.025)		0.125*** (0.042)		0.038 (0.039)		0.025 (0.030)	
<i>Treated</i>	0.018*** (0.004)		−0.030*** (0.003)		0.001 (0.024)		0.004 (0.002)	
<i>Growth Industry</i>	−0.026** (0.012)		0.063** (0.024)		0.061 (0.074)		0.007 (0.026)	
Size and age control	Yes		Yes		Yes		Yes	
State-year fixed effects	Yes		Yes		Yes		Yes	

Table 6. (Continued)

Panel C: Conglomerates: Industry prospects				
Dependent variable	Exit (1)	Entry (2)	Purchase (3)	Sale (4)
Industry-state fixed effects	Yes	Yes	Yes	Yes
Observations	329,000	329,000	329,000	329,000
Adjusted R^2	0.17	0.16	0.02	0.02

Notes. This table estimates a variation of Equation (4) to examine the relation between restructuring activities and plant productivity. The dependent variables are the closure dummy or divestiture dummy. *Treated* is an indicator variable that equals one if a given plant experiences a trade shock in the form of major import tariff reduction in the past three years (t , $t - 1$, or $t - 2$) and zero otherwise. *Bottom TFP Tercile*, *Medium TFP Tercile*, and *Top TFP Tercile* are indicator variables that take the value of one if a plant whose TFP is in the bottom 33%, middle 33% (between 33rd and 67th percentile), or top 33% within a firm-segment (three-digit SIC industry) in a given year, respectively, and is zero otherwise. *Above-median TFP* is an indicator variable that is set to one if a plant whose TFP is in the top 50% within a firm-segment (three-digit SIC industry) in a given year. *Growth Industry* is a dummy variable that is set equal to one if a four-digit SIC industry has a positive and above-median change in its total shipment changes over the next five years. All regressions include plant size and plant age as control variables. Both variables are lagged. All other variables are defined in the appendix. Heteroscedasticity robust standard errors are in parentheses and are corrected for clustering at the industry level.

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

reflects their effort to optimize their operations in order to better cope with increased competition, in which case inefficient plants are more likely closed or divested after the trade shock. On the other hand, if conglomerates' higher propensity to restructure is a manifestation of their underlying agency problems, less efficient plants are more likely retained in the aftermath of the import tariff cuts.

To operationalize this, I divide, for each year, plants within each firm-segment (three-digit SIC) into high-, medium-, and low-productivity terciles. This classification ensures that the restructuring decision reflects conglomerates' choice of plants within their firm boundary.³⁶ The results show that, in periods without import tariff shocks, both conglomerates and multiplant-focused firms are less likely to close down or divest their plants in the high TFP tercile as evident in the negative and significant coefficient on *High TFP Tercile*. However, when affected by trade shocks, conglomerates become even more unlikely to shut down plants in the high TFP tercile, but are more likely to close down and divest plants in the low TFP tercile (columns (1) and (2)). In direct contrast, focused firms affected by trade shocks are more likely to close down plants in the low TFP tercile as well as those in the medium and high TFP terciles, likely pointing to a financial constraint explanation: multiplant-focused firms might be forced to close down their productive plants because of the high wage or technology costs necessary to ensure continued operation. Although the evidence is far from being definitive, it is consistent with the notion that treated conglomerates engage in winner-picking when affected by the trade shock, whereas focused firms affected by the tariff shock are forced to shut down operations even if these plants are very productive.

In panel B, Table 6, I link the relative importance of a plant (*Core*) to the relative productivity of the plant within the conglomerate firm. Specifically, I focus on conglomerates and investigate whether plants of low productivity within the core sector are more or less likely to be closed down or divested after the shock. For these tests, instead of dividing plants' productivity into terciles, I conduct a coarser partitioning by dividing plants into above- and below-median groups to ensure each cell is reasonably populated. The results show that core plants are unconditionally less likely to be closed down (negative coefficients on *Core* in column (1)). Core operations that are productive are less likely to be closed down (-0.030 on *Core* $\times$ *Above-median TFP*), and they are even more likely to be retained (i.e., less likely to be closed) when these productive core plants experience the tariff shock (-0.026 on *Core* $\times$ *Treated* $\times$ *Above-median TFP*).

Finally, in panel C, I repeat the baseline restructuring analyses on conglomerate firms to study whether their restructuring takes into account future industry prospects. Specifically, for each four-digit SIC industry, I follow Maksimovic and Phillips (2008) and adopt a forward-looking approach to compute changes in its total value of shipments over the next five years. I rank industries based on these total shipment changes and code industries that have a positive value and an above-median ranking on this measure *Growth Industry*. Panel C indicates that conglomerates' adjustments reflect their tendency to stay within industries with better growth opportunities as the coefficient on *Treated* $\times$ *Growth Industry* is significantly negative and positive for plant closure (column (1)) and plant opening (column (2)), respectively.

Overall, the results in Table 6 are consistent with Hypothesis 3a: conglomerates seem to pick winners while restructuring. They tend to retain/open plants that are highly productive, and the same phenomenon holds for their structuring within the core segment. Moreover, conglomerates tend to expand into industries with better growth prospects. Taken together, these results suggest that conglomerates' restructuring is more consistent with the winner-picking hypothesis (Hypothesis 3a).

4.3. Quantifying the Effect: Intensive vs. Extensive Margin

Overall, the results so far suggest that, at the intensive margin, conglomerates can shield the negative impact of the import tariff shock by maintaining a relatively higher productivity compared with their focused peers. At the extensive margin, conglomerates are more likely to restructure, and these restructuring activities reflect their preference to refocus on their core competency as well as retain more productive plants. However, it is unclear which margin—intensive or extensive—dominates.

In this section, I try to quantify each channel's economic importance with a matched sample of focused plants that have characteristics similar to the conglomerate plants as described in Section 3.2.4. Conceptually, by comparing the productivity changes around trade shocks between these actual conglomerate plants and the counterfactual consisting of their matched focused peers, I can assess the importance of each adjustment margin. Specifically, I exploit the decomposition method used by Davis et al. (2014),³⁷ which is suitable for my setting. Specifically, let ΔP denote the change in productivity and express the average two-year productivity change as

$$\Delta P_t = \left[S_{t+2}^C P_{t+2}^C - S_t^C P_t^C \right] + \left[S_{t+2}^N P_{t+2}^N - S_t^X P_t^X \right] + \left[S_{t+2}^A P_{t+2}^A - S_t^D P_t^D \right], \quad (5)$$

where S denotes an employment share; P denotes a TFP value; and C , N , X , A , and D denote continuers, entrants, exits, acquisitions, and divestitures, respectively. For instance, P_{t+2}^N is the average TFP of the newly opened plants two years after the trade shock. The average TFP change for controls ($\Delta \tilde{P}_t$) is defined analogously. Equation (6) breaks down the two-year change in firm-level TFP into three distinct parts: the net weighted TFP change by the incumbent plants (continuers), by the portfolio of closed and opened plants (exits and entrants), and by those acquired and divested establishments (acquisitions and divestitures). I combine the two expressions and express all TFP terms as deviations from TFP values for

the control continuers, cancel redundant terms, and rearrange to obtain the following expression:

$$\begin{aligned} \Delta P_t - \Delta \tilde{P}_t = & S_{t+2}^C \left(P_{t+2}^C - \tilde{P}_{t+2}^C \right) - S_t^C \left(P_t^C - \tilde{P}_t^C \right) \\ & + S_{t+2}^N \left(P_{t+2}^N - \tilde{P}_{t+2}^C \right) - \tilde{S}_{t+2}^N \left(\tilde{P}_{t+2}^N - \tilde{P}_{t+2}^C \right) \\ & - S_t^X \left(P_t^X - \tilde{P}_t^C \right) + \tilde{S}_t^X \left(\tilde{P}_t^X - \tilde{P}_t^C \right) \\ & + S_{t+2}^A \left(P_{t+2}^A - \tilde{P}_{t+2}^C \right) - \tilde{S}_{t+2}^A \left(\tilde{P}_{t+2}^A - \tilde{P}_{t+2}^C \right) \\ & - S_t^D \left(P_t^D - \tilde{P}_t^C \right) + \tilde{S}_t^D \left(\tilde{P}_t^D - \tilde{P}_t^C \right). \end{aligned} \quad (6)$$

This expression decomposes into three components the difference in change in TFP of the conglomerate firms relative to their matched focused peers. The top line of the expression captures the contribution of treated-control differences among continuing plants; the second line estimates the change attributed to plant opening and closure; and the third line isolates the adjustment from plant acquisitions and divestitures.³⁸ The employment shares variables can be computed by taking the ratio of the employment of plants by their status to the total employment of the firm. For example, to obtain $\tilde{S}_{t+2}^N$, one can compute the employment share of the newly opened plants at time $t + 2$ for each matched control firm and take the average across all firms. As discussed in Davis et al. (2014), the main advantage of this decomposition is that it combines difference-in-difference estimates with a composition that is often employed in the literature on firm-level productivity dynamics.

I further revise Equation (6) to introduce the element of segment importance. That is, each margin (i.e., continuers, exits, and divestitures) is broken down into those that are in the core segment and those that are in the noncore segment. With this change, the equation becomes the following:

$$\begin{aligned} \Delta P_t - \Delta \tilde{P}_t = & \left[S_{t+2}^{C,Core} \left(P_{t+2}^{C,Core} - \tilde{P}_{t+2}^C \right) - S_t^{C,Core} \left(P_t^{C,Core} - \tilde{P}_t^C \right) \right] \\ & + \left[S_{t+2}^{C,Noncore} \left(P_{t+2}^{C,Noncore} - \tilde{P}_{t+2}^C \right) - S_t^{C,Noncore} \left(P_t^{C,Noncore} - \tilde{P}_t^C \right) \right] \\ & + \left[S_{t+2}^{N,Core} \left(P_{t+2}^{N,Core} - \tilde{P}_{t+2}^C \right) - \tilde{S}_{t+2}^N \left(\tilde{P}_{t+2}^N - \tilde{P}_{t+2}^C \right) \right] \\ & + \left[S_{t+2}^{N,Noncore} \left(P_{t+2}^{N,Noncore} - \tilde{P}_{t+2}^C \right) - \tilde{S}_{t+2}^N \left(\tilde{P}_{t+2}^N - \tilde{P}_{t+2}^C \right) \right] \\ & - \left[S_t^{X,Core} \left(P_t^{X,Core} - \tilde{P}_t^C \right) - \tilde{S}_t^X \left(\tilde{P}_t^X - \tilde{P}_t^C \right) \right] \\ & - \left[S_t^{X,Noncore} \left(P_t^{X,Noncore} - \tilde{P}_t^C \right) - \tilde{S}_t^X \left(\tilde{P}_t^X - \tilde{P}_t^C \right) \right] \\ & + \left[S_{t+2}^{A,Core} \left(P_{t+2}^{A,Core} - \tilde{P}_{t+2}^C \right) - \tilde{S}_{t+2}^A \left(\tilde{P}_{t+2}^A - \tilde{P}_{t+2}^C \right) \right] \\ & + \left[S_{t+2}^{A,Noncore} \left(P_{t+2}^{A,Noncore} - \tilde{P}_{t+2}^C \right) - \tilde{S}_{t+2}^A \left(\tilde{P}_{t+2}^A - \tilde{P}_{t+2}^C \right) \right] \\ & - \left[S_t^{D,Core} \left(P_t^{D,Core} - \tilde{P}_t^C \right) - \tilde{S}_t^D \left(\tilde{P}_t^D - \tilde{P}_t^C \right) \right] \\ & - \left[S_t^{D,Noncore} \left(P_t^{D,Noncore} - \tilde{P}_t^C \right) - \tilde{S}_t^D \left(\tilde{P}_t^D - \tilde{P}_t^C \right) \right]. \end{aligned} \quad (7)$$

Table 7 estimates a difference-in-difference regression in which the dependent variable is the TFP adjusted by the same-year control-continuer productivity, and the independent variables are plant statuses. Remarkably, the estimates in Table 7 are qualitatively the same as the results estimated from the full sample. In particular, as shown in panel A, relative to their matched focused counterparts, conglomerate firms shut down and divest plants of lower TFP, and this effect is present in both the core and noncore plants. For example, TFP of the core and noncore plants closed down by conglomerate firms is, on average, 3.4 and 5.2 log points lower than the control continuers, respectively, and those shut down by focused firms are only 0.5 log points less productive. The difference between the two is large and significant at the 1% level. Panel B concentrates on plant openings and acquisitions. Again, conglomerate firms tend to open more productive plants than their matched focused cohorts, and this is particularly pronounced in the core segment. Panel C, Table 7 exploits the decomposition by applying values from the computed employment shares and TFP changes from panels A and B. Overall, conglomerate firms outperform the focused controls with respect to posttrade shock TFP growth by 2.92 log points.

To put these numbers in perspective, Davis et al. (2014) estimate that the two-year differential change in TFP after private equity buyouts between treated firms and controls is 2.14 log points; in another recent paper, Foster et al. (2016) estimate that reallocation contributes roughly a little over 2 log points per year in industry-level productivity growth. The differential change in TFP of 2.92 log points between treated conglomerates and matched focused firms in this context is, thus, both reasonable and sizeable in magnitude. Summing over terms in the second line of Equation (6), I find that roughly 2.28 log points (out of the 2.92 total) are due to plant opening and closure activities of core and noncore plants, which corresponds to 78% of the differential TFP growth between the treated and control groups. This is the first time, to the best of my knowledge, such decomposition is performed in the context of conglomerate firms, and this result suggests that the reallocation activities in the operation of internal capital market warrants more research effort in the future.

4.4. Aggregate Impact of Conglomerates' Restructuring

In this section, I study the aggregate impact of conglomerates' restructuring at both the firm and industry levels. At the firm level, I focus on how conglomerates' market shares as well as firm sales change in the aftermath of the tariff shock. At the industry level, I study how the relative importance of conglomerate firm's core sectors in a given industry

impact the aggregate productivity after the import tariff shock.

4.4.1. Product Market Consequences. Despite the evidence so far, which shows conglomerates' greater propensity to restructure to focus on their core competency by building more productive capacity, it remains unclear whether these activities translate into better performance in the product market.

In panel A of Table 8, I investigate whether conglomerates that have treated core segments tend to grow their market share and sales faster. For these tests, I aggregate plant-level variables into the firm-segment (three-digit SIC) level. As is evident in column (1), core industries in general have 0.5% higher market share, consistent with prior findings that core sectors are often larger, more efficient segments. The coefficient estimate of 0.002 on *Core* $\times$ *Treated* is positive and significant at the 5% level, whereas the coefficient on *Noncore* $\times$ *Treated* is negative and significant (i.e., -0.002). Column (2) estimates the same specification but uses the logarithm of segment sales as the dependent variable and shows similar results.

In panel B, Table 8, I estimate a similar regression using the full sample of conglomerates and standalones. All observations are aggregated to the firm-year level. The goal of this exercise is to see whether conglomerate firms, because of their restructuring, tend to gain market shares at the expense of their focused peers.³⁹ The negative and significant coefficient of -0.001 on *Treated* indicates that focused firms that are treated by the tariff shocks tend to lose market share. However, conglomerates experiencing trade shocks tend to gain market share (0.004 on *Conglomerate* $\times$ *Treated*).

Overall, the results in this section support the notion that conglomerates' restructuring to shed noncore operations and to focus on core segments lead firms to do better in the product market.

4.4.2. Impact on Aggregate Industry Productivity. This section explores how conglomerates' restructuring impacts aggregate industry productivity. If conglomerates' restructuring activities after the tariff shocks reflect their desire to focus on the core operations presumably by shifting productive resources into these segments, the relative importance of conglomerates' core operations in a given industry may be an important determinant of the overall productivity change at the industry level. I test this conjecture in Table 9. I include as covariates the percentage of conglomerates' core business and the percentage of focused firms in a given industry.

Because these are four-digit SIC industry-year level regressions, I compute the dependent variables using weighted productivity of individual plants, in which

Table 7. Quantification of Different Channels: Intensive vs. Extensive Margins

Panel A: TFP in tariff cut t by plant status in $t + 2$					
Dependent variable: Plant-level log TFP in year t					
Plant status	Conglomerates (1)	Focused control (2)		T-value difference (3)	P-value (4)
Continuers core	0.005 (0.008)	Omitted group	—	0.58	0.291
Continuers noncore	0.033* (0.016)			2.16	0.035
Exiters core	−0.034** (0.015)	Exiters	−0.005 (0.009)	−2.44	0.008
Exiters noncore	−0.052*** (0.007)			−3.12	0.002
Divestitures core	−0.013 (0.010)	Divestitures	−0.007 (0.007)	−0.37	0.356
Divestitures noncore	−0.020*** (0.006)			−1.99	0.025
Industry fixed effects	Yes				
Year fixed effects	Yes				
Observations	445,000				
Panel B: TFP In tariff cut $t + 2$ by plant status in $t + 2$					
Dependent variable: Plant-level log TFP in year $t + 2$					
Plant status	Conglomerates (1)	Focused control (2)		T-value difference (3)	P-value (4)
Continuers core	0.006 (0.005)	Omitted group	—	0.75	0.231
Continuers noncore	0.017* (0.010)			1.72	0.082
Entrants core	0.081*** (0.007)	Entrants	−0.009 (0.013)	4.13	0.000
Entrants noncore	0.039 (0.022)			1.54	0.124
Acquisitions core	0.014 (0.009)	Acquisitions	0.016 (0.016)	−0.21	0.815
Acquisitions noncore	0.020 (0.014)			0.25	0.801
Industry fixed effects	Yes				
Year fixed effects	Yes				
Observations	419,000				
Panel C: Estimated average two-year post tariff change in TFP at conglomerate firms relative to focused controls					
TFP log change differential					2.92
Share of total TFP two-year change differential by margin of adjustment					
Continuing establishments – core					0.05
Continuing establishments – noncore					0.04
Entry and exit – core					0.50
Entry and exit – noncore					0.28
Acquisitions and divestitures – core					0.04
Acquisitions and divestitures – noncore					0.10

Notes. This table presents difference-in-difference regressions based on a matched sample constructed from the main analytical sample described in Section 3.2.4. Cobb–Douglas TFP is estimated using Equation (1). All regressions include industry and year fixed effects as well as plant size and age as controls. Continuers core and noncore are plants that continue operations in the core and noncore industries between time t and $t + 2$, respectively. Exiters core and noncore (divestitures) are plants in the core and noncore industries that exist at t but are shut (divested) at $t + 1$ or $t + 2$, respectively; Entrants core and noncore (acquisitions) are plants in the core and noncore industries that exist at time $t + 2$ but are opened (acquired) at t or $t + 1$, respectively. Core is defined at the three-digit SIC industry level. Heteroscedasticity robust standard errors are in parentheses and are corrected for clustering at the industry level.

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 8. Conglomerates' Restructuring: Product Market Consequences

Panel A		
Dependent variable	Segment market share (1)	Log segment sale (2)
<i>Treated</i> × <i>Core Segment</i> _{<i>t</i>−1}	0.002** (0.001)	0.019** (0.010)
<i>Treated</i> × <i>Noncore Segment</i> _{<i>t</i>−1}	−0.002*** (0.0006)	−0.061*** (0.005)
<i>Core Segment</i> _{<i>t</i>−1}	0.005*** (0.0004)	0.041*** (0.004)
Segment size and age control	Yes	Yes
Firm fixed effects	Yes	Yes
State-year fixed effects	Yes	Yes
Observations	159,000	159,000
Adjusted R ²	0.47	0.90

Panel B		
Dependent variable	Firm market share (1)	Log firm sale (2)
<i>Treated</i>	−0.001*** (0.0003)	−0.023* (0.013)
<i>Treated</i> × <i>Conglomerate</i>	0.004*** (0.0006)	0.098*** (0.007)
<i>Conglomerate</i>	0.022*** (0.0005)	0.686*** (0.006)
Firm size and age control	Yes	Yes
Firm fixed effects	Yes	Yes
State-year fixed effects	Yes	Yes
Observations	520,000	520,000
Adjusted R ²	0.73	0.95

Notes. This table presents segment- or firm-level difference-in-difference regressions in which the dependent variables are market share or logged sales. *Treated* is an indicator variable that equals one if a given plant experiences a trade shock in the form of major import tariff reduction in the past three years (t , $t - 1$, or $t - 2$) and zero otherwise. *Core Segment* is a dummy variable if a three-digit SIC segment accounts for more than 50% of the firm's total value of shipments. *Noncore* is defined as $1 - \text{Core Segment Market Share}$ is the ratio between the sales of a three-digit SIC industry segment within a firm to the sales of the entire three-digit SIC industry in that year. *Log Segment Sale* is the logarithm of the total value of shipments in a three-digit SIC segment. All other variables are defined in the appendix. Heteroscedasticity robust standard errors are in parentheses and are corrected for clustering at the industry level.

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

the weights are plant-level employment in column (1), value of shipments in column (2), and capital stock in column (3). Take column (1) for example: the primary parameter of interest is the interaction term between trade shock and the relative importance of conglomerates' core business in a given industry (*Treated* × % *Emp Conglomerate Core*). The coefficient of 0.033 on this interaction is positive and significant, suggesting that industries for which conglomerates' core business constitute a high percentage

tend to suffer less productivity decline after the trade shocks. I find similar results across various weighting schemes.

4.5. Robustness Tests

Although my key results show that conglomerates affected by the tariff shocks are more likely to restructure, one potential concern may be that the restructuring behavior observed so far is not a result of the import tariff shock, but rather because of some other industry-specific trends or temporary productivity shocks. To rule out these concerns, I create placebo treatments by randomly assigning fake treatments to industries that, in reality, do not experience any major tariff reduction. Table 10 contains the results of this exercise. For none of the four margins do I observe a significantly positive relation between the likelihood of restructuring and the interaction term (*Conglomerate* × *Treated Dummy*). In fact, for plant opening, the coefficient on the interaction term is negative and marginally significant. Therefore, if anything, any potential endogeneity along the lines discussed would bias against finding my results.

Another potential concern is that the main productivity measure used in the paper exploits a simple Cobb–Douglas specification, which imposes a specific structure on the production function. Specifically, there are two possible issues. The first is that the productivity estimation as specified in Equation (1) could be subject to two well-known problems: the selection bias and the simultaneity bias (see Akerberg et al. 2007). To address these two problems at the same time, I reestimate productivity by following the Olley and Pakes (1996) structural approach. Second, ideally for productivity estimation, one would use the physical quantity of the actual goods being produced as the dependent variable in Equation (1). Unfortunately, these data are not available from ASM/CMF. As a result, if conglomerate plants face a different demand compared with focused plants, the main results on productivity (Table 3) could simply be capturing the conglomerates' superior pricing ability as opposed to their actual productivity advantage relative to the focused firms. To mitigate this concern, I follow the approach in Foster et al. (2008) and reestimate the main specification on the subsample of plants producing homogeneous products for which quality and prices are unlikely to vary significantly across different firms. Next, although defining the conglomerate organizational form makes results from this paper more comparable to the existing literature on conglomerates and diversification, there seems to be an apparent misalignment between the definition of conglomerates at the three-digit SIC level and the definition of trade shocks at the four-digit SIC level. Therefore, I also examine the robustness of the

Table 9. Conglomerates' Restructuring: Aggregate Industry Productivity

Dependent variable	Emp Weighted TFP (1)	Sales Weighted TFP (2)	Capital Stock Weighted TFP (3)
<i>Treated</i>	−0.037*** (0.012)	−0.030*** (0.010)	−0.030** (0.012)
<i>Treated</i> × % <i>Emp Conglom. Core</i>	0.030** (0.015)		
% <i>Emp Conglom. Core</i>	−0.004 (0.015)		
<i>Treated</i> × % <i>Emp Focused Firms</i>	0.014 (0.025)		
% <i>Emp Focused Firms</i>	−0.119*** (0.016)		
<i>Treated</i> × % <i>Sales Conglom. Core</i>		0.027** (0.012)	
% <i>Sales Conglom. Core</i>		−0.053*** (0.012)	
<i>Treated</i> × % <i>Sales Focused Firms</i>		0.014 (0.022)	
% <i>Sales Focused Firms</i>		−0.061*** (0.015)	
<i>Treated</i> × % <i>Cap. Stock Conglom. Core</i>			0.033** (0.016)
% <i>Cap. Stock Conglom. Core</i>			−0.010 (0.016)
<i>Treated</i> × % <i>Cap. Stock Focused Firms</i>			0.012 (0.022)
% <i>Cap. Stock Focused Firms</i>			−0.117*** (0.016)
Industry size	0.030*** (0.003)	0.082*** (0.003)	0.034*** (0.003)
Industry fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Adjusted R^2	0.95	0.95	0.95
Observations	11,000	11,000	11,000

Notes. This table presents industry level difference-in-difference regressions in which the dependent variables are industry productivity weighted by employment (column (1)), value of shipments (column (2)), or capital stock (column (3)). *Treated* is an indicator variable that equals one if a given industry experiences a trade shock in the form of major tariff reduction in the past three years (t , $t - 1$, or $t - 2$). % *Emp Conglom. Core* and % *Emp Focused Firms* are the percentage of a four-digit SIC industry's total employment that belongs to conglomerate firms' core businesses and focused firms, respectively. Variables using sales or capital stock are defined analogously and their exact definitions are in the appendix. Heteroscedasticity robust standard errors are in parentheses and are corrected for clustering at the industry level.

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

main findings when conglomerates are defined at the four-digit SIC level.

Finally, TFP estimation places strong assumptions on the form of the production function. To examine whether the main results are robust to other profitability/performance measures free of such assumptions, I reestimate the main regression by employing two more measures: return on capital (ROC) and operating margin (OM). OM is calculated as the value of shipments minus labor and material costs, all divided by shipments. ROC is defined analogously, but uses capital stock as the denominator. In Table 11, I reestimate results from Table 3 using these approaches, samples, and alternative performance and find qualitatively

similar results, which provides additional confidence on the original results.

5. Discussion and Remarks

There are several caveats to keep in mind when interpreting these results: First, although this paper provides consistent evidence that conglomerates seem to be better able to cope with import tariff shocks by restructuring to refocus on their core competency, build more productive plants, and have better product market outcomes in the future, it does not offer any generalizable conclusion on the optimality of organizational form or on the value implications of being a conglomerate (i.e., conglomerate discount versus

Table 10. Falsification Test

Dependent variable	Closure (1)	Opening (2)	Acquisition (3)	Divestiture (4)
<i>Conglomerate Dummy_{t-1}</i>	-0.040*** (0.002)	-0.042*** (0.004)	0.028*** (0.002)	0.025*** (0.002)
<i>Placebo Treated × Conglomerate Dummy_{t-1}</i>	-0.003 (0.008)	0.001 (0.007)	0.000 (0.002)	-0.004** (0.003)
<i>Placebo Treated</i>	0.004 (0.003)	0.001 (0.003)	0.001 (0.002)	-0.002 (0.002)
Size and age control	Yes	Yes	Yes	Yes
Industry-state-year mean	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
State × Year fixed effects	Yes	Yes	Yes	Yes
Observations	871,000	871,000	871,000	871,000
Adjusted <i>R</i> ²	0.15	0.14	0.01	0.01

Notes. This table estimates a variant of the main specification as in Equation (3) using placebo treatment. *Placebo Treated* is an indicator variable that takes a value of one if an industry experiences a placebo tariff cut in year t , $t-1$, or $t-2$. Placebo tariff cuts are randomly assigned to four-digit SIC industries that do not actually experience them in a given year. *Conglomerate Dummy* is an indicator variable that equals one if a firm operates across more than one three-digit SIC industry. All regressions are based on a sample from 1976 to 2004 and include plant size and age as well as industry-state-year mean values of the dependent variable as control variables. Heteroscedasticity robust standard errors are in parentheses and are corrected for clustering at the industry level.

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

premium). In fact, if anything, conglomerate plants are less productive compared with their focused industry peers in the cross-section, which suggests that

conglomerates might be inefficient organizational structures to begin with. In coping with the import tariff shock, however, it seems that conglomerates

Table 11. Robustness Test: Alternative Productivity Estimations

Dependent Variable	TFP (Cobb–Douglas) at four-digit SIC (1)	Homogeneous goods industry only (2)	Structural TFP (OP) (3)	Return on capital (4)	Operating margin (5)
<i>Treated</i>	-0.027** (0.011)	-0.019* (0.010)	-0.007* (0.004)	-0.029*** (0.016)	-0.065** (0.029)
<i>Treated × Conglomerate_{t-1}</i>	0.023** (0.010)	0.020** (0.009)	0.014*** (0.004)	0.025*** (0.009)	0.071*** (0.025)
<i>Conglomerate_{t-1}</i>	0.002 (0.003)	-0.015** (0.007)	-0.022*** (0.002)	-0.016** (0.007)	-0.036* (0.020)
Size and age control	Yes	Yes	Yes	Yes	Yes
Industry-state-year mean	Yes	Yes	Yes	Yes	Yes
Plant fixed effects	Yes	Yes	Yes	Yes	Yes
State × Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	798,000	100,000	871,000	871,000	871,000
Adjusted <i>R</i> ²	0.46	0.48	0.66	0.29	0.30

Notes. This table presents plant-level difference-in-difference regressions as outlined in Equation (2). The dependent variable is TFP of the plant estimated using various methods or samples in columns (1)–(3). Column (1) uses TFP estimated using the Cobb–Douglas production function as in Equation (1). Column (2) uses the same productivity measure but only for a subset of industries producing homogeneous goods as defined in Foster et al. (2008). Column (3) uses a productivity measure structurally estimated using the Olley and Pakes (1996) approach. The dependent variable is operating margin and return on capital in columns (4) and (5), respectively. *Operating Margin* is value of shipments minus labor and material costs, all divided by shipments. *Return on Capital* is value of shipments minus labor and material costs, all divided by capital stock. *Treated* is an indicator variable that equals one if a given plant experiences a trade shock in the form of major import tariff reduction in the past three years ($t-1$, or $t-2$) and zero otherwise. *Conglomerate* is an indicator variable that takes the value of one if a firm operates in more than one three-digit SIC industry in a given year and is zero otherwise. The control variables are plant size and plant age. *Plant Size* is the natural logarithm of the plant's value of shipments. *Plant Age* is the natural logarithm of one plus the number of years since the plant has been in the LBD. Both variables are lagged. Industry-state-year mean is the average value of the dependent variable in a four-digit SIC-state-year. All other variables are defined in the appendix. Heteroscedasticity robust standard errors are in parentheses and are corrected for clustering at the four-digit SIC industry level.

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

correct their mistakes by engaging in more restructuring compared with their focused peers, and these restructuring activities seem to generate positive impact on conglomerates' performance in the product market and aggregate industry productivity. Second, although this paper provides an extensive set of evidence on conglomerates' real actions following import tariff shocks at both the intensive and extensive margins, it is silent on the financing aspect. Specifically, although financial constraint and the lack of access to external capital are likely culprits that prevent focused firms from engaging in productivity-improving restructuring, the paper does not provide direct evidence on how firms' reactions are correlated with characteristics of their balance sheet. Therefore, future work to explore how the interaction between conglomerates' internal capital market and their presumably better access to the external capital market facilitates their response to shocks, including but not limited to import tariff reduction episodes, would be interesting. I acknowledge these limitations and caution readers to keep these in mind when interpreting its results.

6. Conclusion

Do firms' internal organization matter for the impact of international trade? This paper attempts to shed light on this question by exploiting large reductions in import tariffs as a primary setting that significantly heightens the competition faced by domestic firms. By focusing on import-competing firms, this paper documents that firms' internal organization translates into significantly different corporate restructuring activities that result in aggregate productivity changes around episodes of trade liberalization.

Specifically, this paper provides robust evidence that firms operating across different industries (i.e., conglomerates) are significantly more likely to restructure compared with their focused peers: firms that specialize in one single industry. Moreover, these restructuring actions help conglomerates refocus on their core competency, mitigate productivity loss, and improve product market performance.

Conglomerates in the United States have long been criticized by academicians (e.g., Denis et al. 1997) and financial media alike for their inefficiencies. This paper argues that diversified firms could be inefficiently operated on average, but their multi-industry exposure gives them the organizational flexibility to readjust their asset portfolio to better cope with

significant, negative changes in the product market landscape.

In addition, this seemingly simplistic restructuring motive by conglomerate firms has important implications for aggregate industry productivity. Industries dominated by conglomerates' core business experience an insignificant but positive change in aggregate industry productivity, and the other industries suffer from large productivity losses.

As noted in Bernard et al. (2007), "Specialization and reallocation within the firm may turn out to play an important role in enhancing productivity and realizing welfare gains from trade." The evidence in this paper provides strong support for this conjecture. Firms' internal specialization or diversification plays a significant role in shaping the impact of international trade shocks. This paper also suggests that the mechanism via which this interaction happens, at least in this particular context, is through corporate restructuring. Taken together, this paper provides some of the first microevidence documenting how firms' internal organization could help shape the impact of trade liberalization and offers useful policy implications for trade and boundaries of the firm. Understanding how firms of different organizational structure would respond in light of the recent trade war provides an important and rewarding topic for future research.

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Appendix. Variable Definition

Key Dependent Variables

<i>TFP (Cobb–Douglas)</i>	The difference between actual and predicted output. Predicted output is the amount of output a plant is expected to produce for given levels of inputs (i.e., inputs, labor, and capital). Predicted output is computed using a log-linear Cobb–Douglas production function. Specifically, TFP of plant i in year t is the estimated residual from Equation (1).
<i>Plant Opening</i>	An indicator variable that takes the value of one if a plant is in its first year of existence in the LBD and is set to zero otherwise.
<i>Plant Closure</i>	An indicator variable that takes the value of one if a plant is in its last year of existence in the LBD and is set to zero otherwise.
<i>Plant Acquisition</i>	An indicator variable that takes the value of one if a plant is in its first year of existence for firm i in year t but belongs to a different firm in year $t - 1$ and zero otherwise.
<i>Plant Divestiture</i>	An indicator variable that is equal to one if a plant is sold to another firm in year t and zero otherwise.

Key Independent Variables

<i>Treated</i>	An indicator variable if a plant operates in an industry that has experienced a significant import tariff reduction in the last three years (t , $t - 1$, or $t - 2$) and zero otherwise. A significant import tariff reduction happens when the decline in tariff rates in a given four-digit SIC industry in a given year has to exceed three times the median industry change during the sample period.
<i>Conglomerate</i>	An indicator variable that takes the value of one if a firm operates in more than one three-digit SIC industry in a given year and is zero otherwise.
<i>Core (Segment)</i>	An indicator variable that takes the value of one if a three-digit SIC industry constitutes more than 50% of a conglomerate firm's total sales and is equal to zero otherwise.
<i>Noncore (Segment)</i>	The complement of core industry, that is, $Noncore = 1 - Core$.
<i>Other</i>	An indicator variable that equals one if a plant does not experience a trade shock (i.e., $Treated = 0$) in the past three years itself but belongs to a conglomerate firm that has some other plants experiencing the trade shock.

Other Variables

<i>Total Value of Shipments</i>	Plant-level total value of shipments, that is, sales deflated by Bureau of Economic Analysis (BEA) industry-level deflator.
<i>Total Employment</i>	Total number of employees at the plant level.
<i>Plant Size</i>	The logarithm of the total value of shipments.
<i>Plant Age</i>	The number of years since a plant's first appearance in the LBD.
<i>Value of Shipments</i>	Plant-level total value of shipments, that is, sales deflated by BEA industry-level deflator.
<i>Log Number of Conglomerate Firms</i>	The logarithm of the number of conglomerate firms in a four-digit SIC industry in a year.
<i>No. of Firms</i>	The logarithm of the number of firms in a four-digit SIC industry in a year.
<i>Industry Output</i>	The logarithm of the total value of shipments in a four-digit SIC industry.
<i>Industry Herfindahl</i>	The sum of squared percentage of sales in the four-digit SIC industry.
<i>Change in Industry Output</i>	The annual percentage change in industry output, that is, $(Industry Output_t - Industry Output_{t-1}) / Industry Output_{t-1}$
<i>Percentage of Industry Output by Conglomerates</i>	The percentage of total value of shipments in a four-digit SIC industry that are produced by a conglomerate.
<i>Top TFP Tercile</i>	An indicator variable that is set to one if a plant's TFP is in the top 33% within a firm-segment (three-digit SIC industry) in a given year and is zero otherwise.
<i>Medium TFP Tercile</i>	An indicator variable that is set to one if a plant's TFP is in the middle 33% within a firm-segment (three-digit SIC industry) in a given year.
<i>Bottom TFP Tercile</i>	An indicator variable that is set to one if a plant's TFP is in the bottom 33% within a firm-segment (three-digit SIC industry) in a given year.
<i>Above-median TFP</i>	An indicator variable that is set to one if a plant's TFP is in the top 50% within a firm-segment (three-digit SIC industry) in a given year.
<i>Below-median TFP</i>	An indicator variable that is set to one if a plant's TFP is in the bottom 50% within a firm-segment (three-digit SIC industry) in a given year.

(Continued)

<i>Growth Industry</i>	An indicator variable that is set equal to one if a four-digit SIC industry has a positive and above-median change in its total shipment changes over the next five years.
<i>% Emp Focused Firms</i>	The percentage of a four-digit SIC industry's total employment that belongs to focused firms.
<i>% Sales Conglomerate Core</i>	The percentage of a four-digit SIC industry's total sales that belong to conglomerate firms' core businesses.
<i>% Sales Focused Firms</i>	The percentage of a four-digit SIC industry's total sales that belong to focused firms.
<i>% Capital Stock Conglomerate Core</i>	The percentage of a four-digit SIC industry's total capital stock that belongs to conglomerate firms' core businesses.
<i>% Capital Stock Focused Firms</i>	The percentage of a four-digit SIC industry's total capital stock that belongs to focused firms.
<i>Industry Size</i>	The logarithm of the total value of shipments in a four-digit SIC industry.
<i>Segment Market Share</i>	The ratio between the sales of a four-digit SIC industry-segment within a firm to the sales of the entire four-digit SIC industry in that year.
<i>Log Segment Sale</i>	The logarithm of the total value of shipments in a four-digit SIC segment.
<i>Operating Margin</i>	Value of shipments minus labor and material costs, all divided by shipments.
<i>Return on Capital</i>	Value of shipments minus labor and material costs, all divided by capital stock.

Endnotes

¹Figures are based on "Balance of Payments" basis as published in the 2016 census report, which can be found at <https://www.census.gov/foreigntrade/statistics/historical/goods.pdf>.

¹See, for example, Treffer (2004) and Wacziarg and Welch (2008) for the impact of trade liberalization on industry productivity and economic growth, respectively.

²My sample excludes any manufacturing firm for which one or more of its plants report a positive export.

³Note that, throughout this paper, I use the following terms interchangeably: plant closure and plant exit, plant opening and plant entry, plant sale and plant divestiture.

⁴Intensive margin refers to a simple reallocation of resources across existing establishments, that is, the change in actual usage of a given asset portfolio. Theory work (e.g., Stein 1997) on internal capital market tends to assume that this is how it functions.

⁵See Leamer and Levinsohn (1995) for a review.

⁶Some findings of the earlier studies have been challenged by subsequent papers, such as Villalonga (2004) and Campa and Kedia (2002), which explicitly control for the endogeneity of the conglomerate status. In these later studies, the diversification discount either shrinks or turns into a premium. In addition, Whited (2001) and Çolak and Whited (2007) point out the shortcomings of using industry average Tobin's q to proxy for the investment opportunity of conglomerate segments. A recent paper by Tate and Yang (2015) finds that the internal capital market for labor in conglomerate firms creates benefits for workers to the extent that they experience lower wage losses in job switches. Refer to Maksimovic and Phillips (2013) for an up-to-date review of the literature.

⁷Giroud (2013) and Bernstein et al. (2016) use similar settings to examine the impact of headquarter monitoring and venture capital monitoring, respectively.

⁸Conglomerate firms have been shown to exhibit higher sales growth, cash flow, and expenditure on research and development during industry downturns (Gopalan and Xie 2011) and to invest more aggressively relative to their focused counterparts during the recent financial crisis (Kuppuswamy and Villalonga 2015) as well as during periods of heightened competitive pressure brought about by Walmart's entry into the local retail market (Khanna and Tice 2001). In a similar vein, Almeida et al. (2015) find that, by actively transferring resources from low- to high-growth members, Korean business groups sustained a higher investment and profitability in the aftermath of the 1997 Asian financial crisis.

⁹Note that the main TFP measure is a revenue-based productivity measure. Foster et al. (2008) provide a more detailed discussion on the interpretation and limitations of using a revenue-based productivity measure. In robustness checks (Table 11), I also reexamine the main productivity test using a subset of industries that produce homogeneous products.

¹⁰A good example is Alphabet, Inc., which has a core competency in the search engine technology developed by the previously known Google, Inc. At the same time, the multinational conglomerate owns several subsidiaries/segments, such as Calico, DeepMind, GV, CapitalG, X, Google Fiber, Makani, Sidewalk Labs, Verily, Waymo, Wing, and Loon. Up until 2018, Alphabet had been repeatedly criticized as being efficient and profitable only in Google with the other peripheral divisions being only bets (<https://www.engadget.com/2019/04/29/google-alphabet-earnings-q1-2019/>).

¹¹Import tariff shocks can have a differential impact on the productivity of continuing plants that belong to a focused firm or a conglomerate firm for at least two reasons. First, it might take more capable managers to institute more structured practices to minimize the productivity losses induced by the negative cash flow shocks associated with the import tariff reductions. Conglomerates can do this more easily by exploiting their internal (human) capital market (e.g., Tate and Yang 2015), whereas focused firms might find it difficult to recruit talented managers on short notice. More generally, conglomerates can more easily shift around both their capital and labor resources within their firm boundary to more effectively cope with productivity losses. Second, conglomerates might be able to exploit within-firm, across-establishment shipment to lower their materials costs, thus achieving a better mix of production inputs. These within-firm shipments have been found to be of significant value to trade relationships (Atalay et al. 2019).

¹²As discussed earlier, I acknowledge that my approach cannot rule out the possibility that conglomerates can successfully prevent harmful tariff reductions from taking place. It could be that the tariff reductions we see in data are exactly the ones they can expect to fight off. However, this limitation is unlikely to drive my main results for two reasons: First, although the tariff shocks coincide with major historical events, such as the Canada–U.S. trade agreement, their definition is also partly determined by the econometrician by comparing the annual change of tariff rates to median change during the sample period. Second, if all effects documented are a result of anticipation, we should see more preemptive actions, that is, adjustments before the shocks actually take place.

¹³The results are not driven by whether annual tariff changes are compared with the mean or the median.

¹⁴ CMF and ASM together were known as the Longitudinal Research Database prior to 2002.

¹⁵ The rotating panel is formed two years after each census year. For example, the panel for the 1999–2003 period is formed in 1999, two years after the 1997 census. ASM provides, for each plant, sampling probabilities for inclusion into the sample. Although I use weighted regressions to correct for the oversampling of large, relative to small, plants, my results do not depend on the use of sampling weights.

¹⁶ On average, the ASM covers about 50,000 plants each year, and the CMF in census years covers approximately 350,000 establishments.

¹⁷ The sample period ends in 2004 because of this constraint on the tariff data.

¹⁸ The ad valorem tariff implicitly assumes that the transportation costs associated with the imports count toward the total value of the imports.

¹⁹ The sample ends in 2004 because when this project first started, the last year of available import tariff data are 2002. I end the sample two years after the last import tariff shocks.

²⁰ Note that these filtering criteria have been used in prior literature. For instance, the processing of import tariff data follows Fresard and Valta (2016); the exclusion of administrative records and dropping plants with less than two years of continuous data follows Maksimovic et al. (2011) and Maksimovic and Phillips (2001, 2002, 2008).

²¹ Giroud and Mueller (2015) use a similar restriction criterion but limit their sample to public firms. The results are not affected by this criterion.

²² For example, an observation that has probability one in three of being included in a sample has probability weight equal to three. The results are qualitatively the same if the weights are not applied. Bai et al. (2018), for instance, use a weighting scheme in their empirical examination.

²³ Census policy requires that the number of observations across all the tables be rounded to the nearest thousand.

²⁴ The census only allows disclosure on mean and standard deviation of variables. Disclosure of statistics such as 25% (75% percentile) or median values are prohibited.

²⁵ If trade shocks lead firms to employ capital and labor in different ways than before the shock, it is incorporated in the productivity estimation. In addition, estimating Equation (1) for each year eliminates any contamination concerns, that is, pretreatment parameters estimated using posttreatment variables.

²⁶ Levinsohn and Petrin (2003) is a seminal paper that addresses the simultaneity problem by using intermediate inputs to proxy for the productivity shock. Although the results in this paper still hold using this method, the approach is not particularly suitable for the purpose of this paper. In their paper, half of the plants in their sample report zero capital investment, which does not apply to the sample used in this paper.

²⁷ The results in Figure 4 are capital-weighted average TFP. In unreported results, I also try equal weighting and weighting using employment as well as aggregating the plant-level productivity to the four-digit SIC segment level. The graphs look similar in all cases.

²⁸ The results in Figure 4 are capital-weighted average TFP. In unreported results, I also try equal weighting and weighting using employment, as well as aggregating the plant level productivity to the four-digit SIC segment level. The graphs look similar in all cases.

²⁹ Even at the three-digit SIC, the industry-state-year fixed effects would require estimation of 210,250 fixed effects ($145 \times 3\text{-digit SIC} \times 29 \text{ years} \times 50 \text{ states} = 210,250$).

³⁰ Gormley and Matsa (2014) advise that fixed effects should be used in place of including the average value at a certain level to minimize bias. Our results are unchanged if I omit these average values of the dependent variables from the list of control variables.

³¹ Note that the inclusion of plant fixed effects means that β_2 is identified from “switching plants,” which are plants that have belonged to conglomerate firms at one time and focused firms at another.

³² This result contrasts to Schoar (2002), which finds that conglomerate plants become more productive after diversification actions for two reasons: First, Schoar (2002) studies only public firms. In addition, the original specification in Schoar (2002) includes only plant fixed effects with no year fixed effects. In unreported results, I verify that these two reasons contribute to the differences between the negative coefficient on *Conglomerate* and the positive effect in table 3 of Schoar (2002).

³³ Linear probability models provide a consistent and less biased estimate when multiple fixed effects are included as suggested in Lancaster (2000) and Baltagi (2008). Nonlinear models often suffer from incidental parameter bias.

³⁴ LBD starts from 1976 and ends in 2012 when this project was first started. To deal with the truncation problem, the “Opening” dummy is only defined from 1977 onward.

³⁵ Despite the statistical significance, the economic magnitude is relatively small.

³⁶ Note that, for focused firms, this partitioning is essentially a firm-year partitioning as, by definition, these firms operate only in one three-digit SIC industry.

³⁷ Davis et al. (2014) examine the impact of private equity on firms’ efficiency, employment, and investment decisions focusing on a short-term window. This makes their methodology particularly suitable for my purpose.

³⁸ Note that the employment shares variables depend on the plant status. For example, for plants that are opened during the two-year horizon, I use their employment share at $t + 2$, the end of the horizon. For plants that are closed, I use their employment share at t , the beginning of the horizon. (See Foster et al. 2001 for a detailed discussion.)

³⁹ For conglomerates, we calculate the dependent variable based on their largest industry.

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